Path Integrals

Part II: Relativistic Quantum Mechanics

The worldline formalism and the Klein–Gordon field

Typeset from handwritten notes (pp. 1–40), corrected, reorganized and expanded

Continues Part I (non-relativistic quantum mechanics) and leads into Part III (quantum field theory)

October 2026

Preface

These notes are a typeset version of handwritten lecture notes on path integrals for a relativistic particle (handwritten pp. 1–40). They are Part II of a series. Part I, *Path Integrals in Quantum Mechanics*, treats the non-relativistic theory; Part III treats quantum field theory. Chapter, equation and figure numbers continue those of Part I, so Chapters 1–9 are Part I and this part begins at Chapter 10. A reference such as eq. (6.7) or §8.7 therefore points into Part I, and appendices here continue from Appendix C.

Three kinds of changes were made, as in Part I.

1. **Corrections.** Errors of physics, mathematics and units have been fixed in place. The complete log, page by page, is in Appendix D.
2. **Reordering.** The Klein–Gordon material (handwritten pp. 19–24) now comes before the comparison with the worldline theory, which needs it. A long Fourier-series argument for gauge fixing (pp. 13–16) is replaced by a two-line argument and by a proper Faddeev–Popov calculation. Additional Note 03 (pp. 37–40) becomes the final chapter, because it is the bridge to Part III.
3. **Additions**, marked ★ in headings: the Hamiltonian form of the worldline theory and the Klein–Gordon equation as a constraint; the Faddeev–Popov measure; the role of $i\eta$ and of the range of the proper-time integral; the Euclidean propagator, the classical limit and the non-relativistic limit as checks; antiparticles and the limits of a one-particle theory; Feynman versus retarded propagation; gauge covariance, the Aharonov–Bohm phase and the seagull vertex; vacuum loops; exact Feynman rules for a cubic interaction; the vacuum wave functional; worked solutions to all exercises.

Tags such as [hw. p. 12] point to the handwritten relativistic notes.

Conventions used throughout Part II

- Mostly-plus signature: $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, \ldots, +1)$, so $x^2 = \eta_{\mu\nu} x^\mu x^\nu = -t^2 + \mathbf{x}^2$. Spacetime has D dimensions and $N = D - 1$ space dimensions. Proper time is $\mathrm{d}\tau^2 = -\eta_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu$.
- $c = 1$ and $\hbar$ is kept explicit. A particle of mass m has inverse (reduced) Compton wavelength $\mu \equiv m/\hbar$. On the field side, plane waves are $e^{ik\cdot x}$ with wave vector k^μ and $\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + \mu^2}$.
- The worldline parameter is $s \in [0, 1]$, the einbein is $\lambda(s) > 0$, and the proper-time length (the modulus) is $T \equiv \int_0^1 \lambda \, \mathrm{d}s$.
- The regulator is written $i\eta$ with $\eta \to 0^+$, as in Part I; ε is a time step. Euclidean

time is $x^0 = -i\bar{x}^0$, the same branch as $t = -iu$ in Part I.

- $D(x-y) = \langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle$ (Wightman function), $G_{\rm F}(x-y) = \langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle$ (Feynman), $\Delta \equiv iG_{\rm F}$ (the handwritten definition), and $G_{\rm R}$ is the retarded function.
- Peskin & Schroeder use $(+,-,-,-)$ and $\hbar = 1$. Then $k^2_{\rm Peskin} = -k^2_{\rm here}$ and $\dfrac{-i}{k^2+\mu^2-i\eta}\Big|_{\rm here} = \dfrac{i}{k^2-\mu^2+i\eta}\Big|_{\rm Peskin}$.

quantity	handwritten	here
proper-time length (modulus)	e	T
mass parameter on the field side	m	$\mu = m/\hbar$
electric charge	e	q
coupling constants	λ, λ_3, λ_4	g_3, g_4
wave vector on the field side	p	k
regulator	$i\epsilon$	$i\eta$
einbein, normalization constants	$\lambda(s)$, α, ν	unchanged

What carries over from Part I

- The Fresnel integral and stationary phase (§2.5): every Gaussian integral here, and the classical limit of the proper-time integral (§13.8).
- $S[x]$ versus $S_{\rm cl}$ and Hamilton–Jacobi theory (§3.2–§3.4): for a free relativistic particle $S_{\rm cl} = -m\tau$.
- The phase-space path integral (§5.2) and the warning of §5.5 that the Legendre transform fails for constrained systems. The relativistic particle is the first such system (§11.6).
- The free-particle normalization (§9.4): the x-integrals of Chapter 13.
- The retarded Green's function and its energy representation (§6): the relativistic propagator is $\hat{G}_r(E)$ of eq. (6.7) at $E = 0$ (§14.2).
- Insertions, time ordering and contact terms (§7): the seagull vertex (§17.5).
- Complex time, the heat kernel and the partition function (§8): the Euclidean propagator (§13.7) and closed worldlines (§17.7).
- The oscillator, its modes and determinants (§9): the field as oscillators (Chapter 15), the gauge-fixing measure (§12.6) and the vacuum wave functional (§18.4).

idea	Part I	Part II	Part III
time slicing and the measure	§4.4, §5.3, §9.4	§13.3, §18.5	the field measure $\mathcal{D}\phi$, §19.3, §19.4
stationary phase	§2.5, §4.6	§13.8	the loop ($\hbar$) expansion, §20.5, §26.5
Green's functions and $i\eta$	§6, §8.9	§13.4–§13.6, §14.2, §15.7	propagators from $Z[J]$, §23.4, §24.1
insertions and contact terms	§7	§17.3–§17.5	Schwinger–Dyson equations, Ch. 28
Euclidean time	§8	§13.7, §15.5	Euclidean field theory, §23.5, Ch. 25
closed paths	§8.11	§17.7	the zero-point energy and the one-loop effective action, §24.4, Ch. 29
oscillators and determinants	§9	§12.6, Ch. 15, §18.4	Gaussian functional integrals, Ch. 22, §24.1
constraints and gauge fixing	§5.5	Ch. 11–12, §14.2	ghosts for gauge fields: beyond this volume

Contents

Preface **1**

10 From paths to worldlines **6**
10.1 What must be generalized 6
10.2 Four ways to do quantum mechanics 6
10.3 Worldlines and their parametrization 6
10.4 Proper time and geodesics 7

11 Worldline actions **10**
11.1 The square-root action 10
11.2 Equations of motion, and the massless case 10
11.3 Why the square root is a problem 11
11.4 The einbein action 11
11.5 Reparametrization invariance 12

11.6 Hamiltonian form: the mass-shell constraint★ 13

12 Reparametrization as a gauge symmetry **15**
12.1 Physical and gauge symmetries 15
12.2 Dividing by the gauge volume 15
12.3 Orbits and slices 16
12.4 The proper-time length: the only invariant 16
12.5 Why not fix the gauge on x? 18
12.6 The Faddeev–Popov measure★ 18

13 The free relativistic propagator **21**
13.1 The gauge-fixed path integral 21
13.2 Proper-time gauge 21
13.3 Evaluating the Gaussian 21
13.4 Convergence and the $i\eta$ prescription★ 22
13.5 Momentum space and Schwinger's representation★ 23
13.6 Which Green's function? The range of T★ 23
13.7 The Euclidean propagator: Brownian motion in spacetime★ 24
13.8 Two checks: the classical and the non-relativistic limits★ 26

14 The propagator as a Green's function, again **28**
14.1 The question 28
14.2 The worldline Hamiltonian and its resolvent★ 28
14.3 First quantization: the Klein–Gordon equation as a constraint★ 29
14.4 Negative frequencies and antiparticles★ 30

15 The Klein–Gordon field **31**
15.1 Each mode is an oscillator 31
15.2 Mode expansion 32
15.3 Green's functions 32
15.4 Energy and momentum representations 32
15.5 Euclidean version 33
15.6 One-particle states and the limits of localization 33
15.7 What "causal" means: Feynman versus retarded★ 34

16 Worldline equals field theory **36**
16.1 The question 36
16.2 Method I: the amplitude from the field side 36
16.3 Method II: explicit calculation 36
16.4 The result 37
16.5 What the equivalence means 38
16.6 Why the worldline description is universal 38

17 Diagrams, background fields and interactions **39**
17.1 A line is a worldline path integral . . . 39
17.2 Insertions . . . 39
17.3 A charged particle in a background field: scalar QED . . . 39
17.4 Gauge covariance and the Aharonov–Bohm phase★ . . . 41
17.5 The seagull vertex from a contact term★ . . . 41
17.6 Free N-point functions and Wick's theorem . . . 41
17.7 Closed worldlines: the vacuum loop★ . . . 42
17.8 Interactions from vertices . . . 43
17.9 Summary: a theory is a propagator plus a topology . . . 45

18 From particles to fields: the wave-functional picture **46**
18.1 Two analogies, not one . . . 46
18.2 Field eigenstates . . . 47
18.3 The momentum as a functional derivative . . . 47
18.4 The vacuum wave functional★ . . . 48
18.5 Toward Part III: the field path integral★ . . . 48

C Solutions to the exercises **50**
C.1 The oscillator Green's function . . . 50
C.2 The Klein–Gordon Green's function . . . 50
C.3 The energy and momentum representations . . . 50
C.4 The Euclidean representation . . . 51
C.5 Lorentz invariance of the normalization . . . 51
C.6 The commutator in momentum space . . . 51
C.7 The retarded Green's function . . . 51

D Map of the handwritten notes and correction log **52**
D.1 Where each handwritten page went . . . 52
D.2 Corrections . . . 52

10 From paths to worldlines

Part I was built on a single formula, eq. (4.11):

$$\langle x_f, t_f | x_i, t_i \rangle = \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x \; e^{iS_{if}[x]/\hbar}, \qquad S_{if}[x] = \int_{t_i}^{t_f} \mathrm{d}t \, L(x, \dot{x}, t). \tag{10.1}$$

This chapter asks what has to change when the particle is relativistic, and sets up the language of worldlines.

10.1 What must be generalized [hw. pp. 1–2]

Read relativistically, the labels of (10.1) are already natural: (x_i, t_i) is the initial *event* and (x_f, t_f) the final one. Two things are not relativistic.

- **The paths.** A non-relativistic path is the graph $x(t)$ of a function of time. Time is the parameter and position the variable.
- **The action.** $S_{if} = \int \mathrm{d}t \, L$ singles out the time coordinate.

The asymmetry runs deeper than notation. In non-relativistic quantum mechanics $\hat{x}$ is an operator, a dynamical variable, while t is a parameter that labels states. Treating space and time alike leaves two options: promote t to a variable, as x already is, or demote x to a label, as t already is. The first option leads to the *worldline* formalism of this part. The second leads to *field theory*, where x and t both label an operator $\hat{\phi}(t, \mathbf{x})$. The two routes meet in Chapter 16.

10.2 Four ways to do quantum mechanics [hw. pp. 19–20]

The handwritten notes summarize the landscape with two adjectives. An approach is a *worldline* (WL) approach if it sums over the paths of a particle, and a *spacetime* (ST) approach if its basic objects are operators or fields living on spacetime. Figure 21 shows how the four combinations are related.

10.3 Worldlines and their parametrization [hw. pp. 1–2]

A relativistic path is a curve $x^\mu(s)$ in Minkowski space, $\mu = 0, 1, \ldots, N$, traced out by a parameter $s \in [0, 1]$ with $x(0) = x_i$ and $x(1) = x_f$. The amplitude to be constructed is

$$\langle x_f | x_i \rangle = \int_{x(0)=x_i}^{x(1)=x_f} \mathcal{D}x \; e^{iS_{if}[x]/\hbar}, \qquad S_{if}[x] = \int_0^1 \mathrm{d}s \, L(x, \dot{x}), \qquad \dot{x}^\mu \equiv \frac{\mathrm{d}x^\mu}{\mathrm{d}s}. \tag{10.2}$$

The Lagrangian may not depend on s explicitly; that will be forced by reparametrization invariance (§11.5).

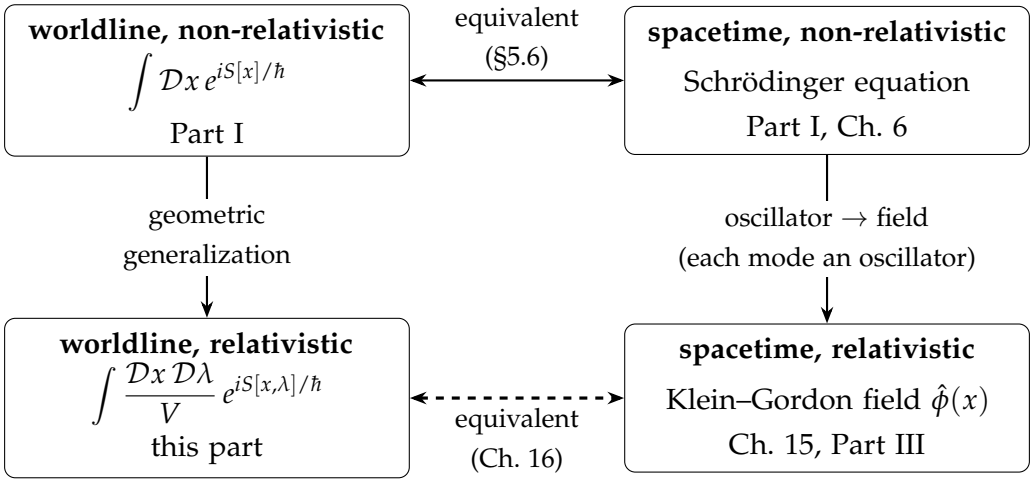


Figure 21: The four combinations of worldline/spacetime and non-relativistic/relativistic [hw. p. 20]. The horizontal arrows are equivalences; the lower one is the main result of this part. The vertical arrows are the generalizations used to build the relativistic theories.

Along a physical, timelike path the proper time τ increases monotonically, so τ is a good parameter, and so is any $s = s(\tau)$ with $\mathrm{d}s/\mathrm{d}\tau > 0$. Which one we use has no physical meaning: the same worldline can be traced quickly or slowly. That is why the state labels in (10.2) carry no s, in contrast with the t in (10.1) [hw. p. 2]. In non-relativistic quantum mechanics t is physical and appears in the state; here s is not and does not.

> **Caution: the paths in the sum are not all timelike**
>
> Classical worldlines are timelike, but the path integral sums over *all* curves from x_i to x_f (Fig. 22). Most of them contain spacelike pieces, and many run backward in x^0 for a while. Proper time cannot parametrize such curves, which is why a generic parameter s is used. The backward-running pieces are not a nuisance: they describe antiparticles (§14.4).

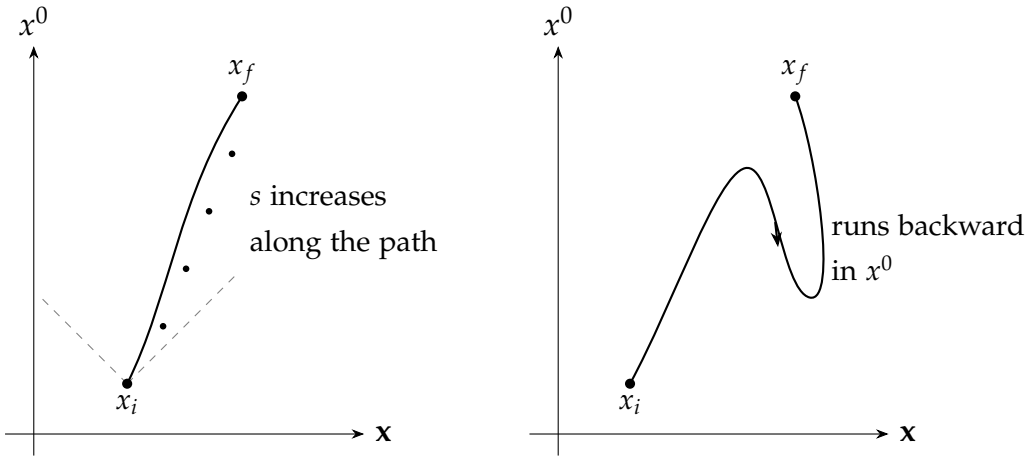
10.4 Proper time and geodesics [hw. p. 3]

With $c = 1$ and the mostly-plus metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, \ldots, +1)$, the proper time along a timelike curve is

$$\mathrm{d}\tau^2 = -\eta_{\mu\nu}\, \mathrm{d}x^\mu \mathrm{d}x^\nu = \mathrm{d}t^2 - \mathrm{d}\mathbf{x}^2. \tag{10.3}$$

A free massive particle moves on a straight line, $\mathrm{d}^2x^\mu/\mathrm{d}\tau^2 = 0$. A free massless particle moves on a straight *null* line, along which $\mathrm{d}\tau = 0$. Its equation of motion needs a parameter other than τ (§11.4).

Straight lines are the geodesics of Minkowski space, and for timelike curves the



(a) a timelike worldline (b) a path that turns back in time

Figure 22: (a) A classical worldline is timelike: it stays inside the light cone (dashed) of every one of its points, and proper time can serve as its parameter. (b) The path integral also contains curves that turn back in time. In §14.4 the backward segment is read as an antiparticle.

geodesic is an extremum of proper time. It is a *maximum*, not a minimum. In the rest frame of the straight line between two events, the straight line has $\tau = \Delta t$, while any other curve has $d\tau = \sqrt{dt^2 - d\mathbf{x}^2} \leq dt$ at every step. A zig-zag of nearly null segments can have arbitrarily small proper time, so there is no minimum at all (Fig. 23). This is the twin paradox.

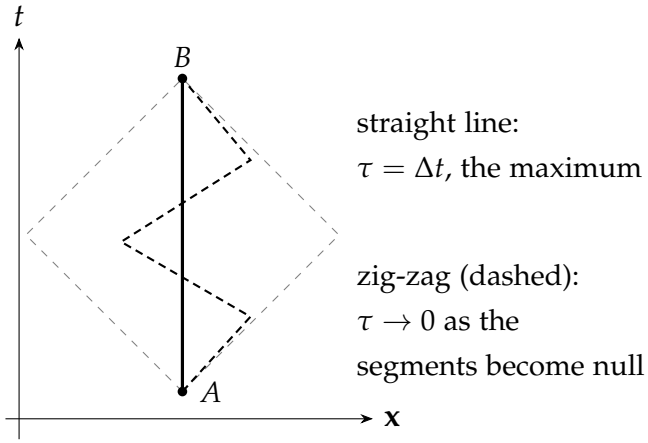


Figure 23: Among timelike curves from A to B the straight line has the largest proper time. Curves close to the light cone (gray dashed) have almost none, so there is no minimum.

Caution: extremum means maximum here

The handwritten notes describe a geodesic as a path of extremal length, "max or min" [hw. p. 3]. For timelike curves in Minkowski space it is always a maximum of τ. The action built from it in the next chapter, $-m\int \mathrm{d}\tau$, is therefore a genuine minimum, so for the free relativistic particle "least action" is literally true. In curved spacetime a geodesic stops being a maximum after it passes a conjugate point. This is the same phenomenon as the oscillator of §9.6, and Part I's warning (§3) applies again.

11 Worldline actions

11.1 The square-root action [hw. pp. 3–4, 7–8]

The free particle extremizes proper time (§10), so the natural candidate for the action is proper time itself:

$$S^\circ[x] = -m\int \mathrm{d}\tau = -m\int_0^1 \mathrm{d}s\,\sqrt{-\eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}, \qquad x(0)=x_i, \quad x(1)=x_f. \tag{11.1}$$

The handwritten notes write the coefficient as ν and note that it has the dimension of a mass [hw. p. 3]. That it *is* the mass is clearest from the non-relativistic limit. Parametrizing a timelike path by t,

$$S^\circ = -m\int \mathrm{d}t\,\sqrt{1-\mathbf{v}^2} = \int \mathrm{d}t\Big(-m+\tfrac{1}{2}m\mathbf{v}^2+\tfrac{1}{8}m\mathbf{v}^4+\cdots\Big). \tag{11.2}$$

The second term is the free Lagrangian of Part I. The first is the rest energy, which will reappear as the phase $e^{-imc^2t/\hbar}$ (§13.8). The rest are relativistic corrections.

Why the minus sign. There are two reasons, and they are the same reason.

- The straight line *maximizes* τ (Fig. 23), so $-m\int\mathrm{d}\tau$ is smallest on the classical path.
- In Euclidean time, $\tau=-iu$ (the branch $t=-iu$ of Part I, §8.8) [hw. p. 4],

$$\begin{aligned} \frac{i}{\hbar}S^\circ &= -\frac{im}{\hbar}\int(-i\,\mathrm{d}u) = -\frac{m}{\hbar}\int\mathrm{d}u, \\ e^{iS^\circ/\hbar} &= e^{-S_\mathrm{E}/\hbar}, \qquad S_\mathrm{E} = m\int\mathrm{d}u \;\geq 0. \end{aligned} \tag{11.3}$$

  The Euclidean action is m times the Euclidean length of the path. Long paths are suppressed, as in Part I. With the opposite sign the weight would grow with length and the integral would not exist.

The scale of quantum effects. The phase $e^{iS^\circ/\hbar}$ advances by $m/\hbar$ per unit proper time, the Compton angular frequency. Classically m drops out of free motion, since the straight line is the same for every mass [hw. p. 7]. Quantum mechanically, larger m means faster phases and stronger cancellation away from the classical path, so quantum effects are confined to scales below the Compton wavelength $\hbar/mc$ [hw. p. 8].

11.2 Equations of motion, and the massless case [hw. pp. 4, 7]

Varying (11.1) with $\delta x(0)=\delta x(1)=0$,

$$\delta S^\circ = m\int_0^1 \mathrm{d}s\,\frac{\eta_{\mu\nu}\dot{x}^\mu\,\delta\dot{x}^\nu}{\sqrt{-\dot{x}^2}} = -m\int_0^1 \mathrm{d}s\,\frac{\mathrm{d}}{\mathrm{d}s}\Big[\frac{\dot{x}_\nu}{\sqrt{-\dot{x}^2}}\Big]\delta x^\nu. \tag{11.4}$$

For a timelike path $\sqrt{-\dot{x}^2} = \mathrm{d}\tau/\mathrm{d}s > 0$ and $\dot{x}^\mu/\sqrt{-\dot{x}^2} = \mathrm{d}x^\mu/\mathrm{d}\tau$, so

$$\frac{\mathrm{d}}{\mathrm{d}s}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \frac{\mathrm{d}\tau}{\mathrm{d}s}\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} = 0. \tag{11.5}$$

Caution: the square-root action has no massless limit

The handwritten notes read a second branch off (11.5), "$\mathrm{d}\tau/\mathrm{d}s = 0$: massless" [hw. pp. 4, 7]. That branch is not legitimate.

- The derivation divides by $\sqrt{-\dot{x}^2}$, which vanishes on a null path. The square root is not differentiable at zero, so the variation is undefined there.
- At $m = 0$ the action (11.1) vanishes identically and has no equations of motion at all.
- $\mathrm{d}\tau = 0$ is not enough to make a path straight. The null helix $x^\mu(s) = \left(s, R\cos\frac{s}{R}, R\sin\frac{s}{R}, 0\right)$ has $\dot{x}^2 = -1 + \sin^2\frac{s}{R} + \cos^2\frac{s}{R} = 0$ everywhere and is not a geodesic.

Massless particles need the einbein action of §11.4.

11.3 Why the square root is a problem [hw. p. 4]

The handwritten notes flag the square root as a difficulty for the path integral. There are four distinct problems.

1. It is not quadratic in x, so the path integral is not Gaussian. In Part I the free particle and the oscillator were solvable exactly because their actions were quadratic (remark after eq. (2.16)).
2. On a spacelike segment $-\dot{x}^2 < 0$ and the square root is imaginary. The weight $e^{iS/\hbar}$ is then no longer a phase, and the sum does contain such segments.
3. It cannot describe massless particles.
4. Its canonical Hamiltonian vanishes identically (§11.6), so the derivation of Part I, Chapter 5, which starts from $\hat{H}$, has nothing to start from.

11.4 The einbein action [hw. pp. 5, 7, 9]

Different actions can give the same classical equations of motion [hw. p. 5]. Is there one that is quadratic in x? Yes:

$$\boxed{S[x,\lambda] = \frac{1}{\alpha}\int_0^1 \mathrm{d}s\left[\lambda(s)^{-1}\eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu - \lambda(s)\,\nu^2\right], \qquad \alpha,\nu > 0.} \tag{11.6}$$

(The handwritten notes use $\alpha = 2$ on p. 5 and a general α from p. 7 on.) The new variable $\lambda(s)$ is to be integrated over as well, but it appears without derivatives. It is an auxiliary, non-dynamical variable, and its equation of motion is algebraic, a constraint:

$$\frac{\delta S}{\delta \lambda(s)} = \frac{1}{\alpha}\Big(-\frac{\dot{x}^2}{\lambda^2} - \nu^2 \Big) = 0 \quad \Longrightarrow \quad \lambda = \frac{\sqrt{-\dot{x}^2}}{\nu}. \tag{11.7}$$

The positive root is the one that makes $\lambda\, \mathrm{d}s$ a positive length (§12.4). Substituting it back,

$$S = \frac{1}{\alpha}\int_0^1 \mathrm{d}s \left[\frac{\nu\, \dot{x}^2}{\sqrt{-\dot{x}^2}} - \nu\sqrt{-\dot{x}^2} \right] = -\frac{2\nu}{\alpha}\int_0^1 \mathrm{d}s\, \sqrt{-\dot{x}^2} = S^\circ \quad \text{with} \quad \boxed{m = \frac{2\nu}{\alpha}} \tag{11.8}$$

[hw. pp. 5, 7]. The equation of motion for x is $\frac{\mathrm{d}}{\mathrm{d}s}\big(\lambda^{-1}\dot{x}^\mu\big) = 0$. With (11.7) it is the same as (11.5), so the two actions describe the same classical particle.

> **Intuition: the einbein is a ruler**
>
> On the classical path $\lambda\, \mathrm{d}s = \mathrm{d}\tau/\nu$: the einbein converts parameter length into proper time. It is the square root of a metric on the worldline, $g_{ss} = \lambda^2$; "einbein" is German for "one leg", the one-dimensional version of a vielbein. Eliminating λ is a small exercise in the inequality between arithmetic and geometric means. For $-\dot{x}^2 > 0$,
>
> $$\frac{1}{\alpha}\Big(\frac{\dot{x}^2}{\lambda} - \lambda\nu^2\Big) = -\frac{1}{\alpha}\Big(\frac{-\dot{x}^2}{\lambda} + \lambda\nu^2\Big) \le -\frac{2\nu}{\alpha}\sqrt{-\dot{x}^2}, \tag{11.9}$$
>
> with equality exactly at (11.7). The einbein action does the work of the square root by being optimized over the ruler.

The massless case now works. At $\nu = 0$, $S = \frac{1}{\alpha}\int \lambda^{-1}\dot{x}^2$. The λ equation gives $\dot{x}^2 = 0$: the path is null. The x equation gives $\frac{\mathrm{d}}{\mathrm{d}s}(\lambda^{-1}\dot{x}) = 0$, which in the gauge $\lambda = \text{const}$ is $\ddot{x}^\mu = 0$: a straight null line. The helix of the caution above is excluded, since it has $\ddot{x} \neq 0$.

> **Caution: why λ is introduced**
>
> The handwritten notes say that new fields must be introduced "to make the invariance manifest" [hw. p. 9]. The square-root action is already manifestly reparametrization invariant (next section). What the einbein buys is an action that is at once invariant, quadratic in x, and valid at $m = 0$. String theory makes the same move, from the Nambu–Goto area to the Polyakov action with a metric on the world-sheet.

11.5 Reparametrization invariance [hw. pp. 6–7]

A reparametrization is a change of parameter $s \to \tilde{s} = f(s)$ with

$$f(0) = 0, \qquad f(1) = 1, \qquad \dot{f}(s) > 0, \tag{11.10}$$

so that the endpoints and the direction of the path are kept. The coordinates are scalars on the worldline: $\tilde{x}^\mu(\tilde{s}) = x^\mu(s)$, the same point of spacetime with a new label.

For the square-root action,

$$S^\circ[\tilde{x}] = -m\int \mathrm{d}\tilde{s}\,\sqrt{-\Big(\frac{\mathrm{d}\tilde{x}}{\mathrm{d}\tilde{s}}\Big)^2} = -m\int \mathrm{d}s\,\dot{f}\,\sqrt{-\frac{\dot{x}^2}{\dot{f}^2}} = S^\circ[x] \qquad \text{[hw. p. 6].} \tag{11.11}$$

For the einbein action,

$$S[\tilde{x},\tilde{\lambda}] = \frac{1}{\alpha}\int \mathrm{d}\tilde{s}\,\Big[\tilde{\lambda}^{-1}\Big(\frac{\mathrm{d}\tilde{x}}{\mathrm{d}\tilde{s}}\Big)^2 - \tilde{\lambda}\nu^2\Big] = \frac{1}{\alpha}\int \mathrm{d}s\,\Big[\big(\dot{f}\tilde{\lambda}\big)^{-1}\dot{x}^2 - \big(\dot{f}\tilde{\lambda}\big)\nu^2\Big], \tag{11.12}$$

which equals $S[x,\lambda]$ if and only if

$$\boxed{\tilde{\lambda}(\tilde{s}) = \frac{\lambda(s)}{\dot{f}(s)}, \qquad \text{i.e.} \qquad \tilde{\lambda}(\tilde{s})\,\mathrm{d}\tilde{s} = \lambda(s)\,\mathrm{d}s} \tag{11.13}$$

[hw. p. 7]. The einbein transforms like a length element; $\lambda\,\mathrm{d}s$ is invariant. (The handwritten line on p. 6 drops the factor $\frac{1}{2}$ of p. 5; the overall factor is irrelevant here.) For an infinitesimal reparametrization $f(s) = s + \xi(s)$, with $\xi(0) = \xi(1) = 0$,

$$\delta x^\mu = -\xi\,\dot{x}^\mu, \qquad \delta\lambda = -\frac{\mathrm{d}}{\mathrm{d}s}\big(\xi\lambda\big). \tag{11.14}$$

Remark 11.1. An action $\int \mathrm{d}s\,L(x,\dot{x})$ is invariant without an einbein only if L has no explicit s-dependence and is homogeneous of degree one in the velocity, $L(x,c\dot{x}) = c\,L(x,\dot{x})$ for $c>0$. Built from $\dot{x}^2$ alone, the only such Lorentz scalar is the square root. A term $A_\mu(x)\dot{x}^\mu$ is also homogeneous of degree one, which is why the coupling to a vector potential needs no einbein (§17.3).

11.6 Hamiltonian form: the mass-shell constraint★

Square-root action. The canonical momentum is the four-momentum,

$$p_\mu = \frac{\partial L}{\partial\dot{x}^\mu} = \frac{m\,\dot{x}_\mu}{\sqrt{-\dot{x}^2}} = m\frac{\mathrm{d}x_\mu}{\mathrm{d}\tau}. \tag{11.15}$$

Two things happen that never happened in Part I. First, $p^2 = m^2\dot{x}^2/(-\dot{x}^2) = -m^2$ *identically*: the momenta are not independent. Second, the canonical Hamiltonian vanishes,

$$H_\mathrm{c} = p_\mu\dot{x}^\mu - L = -m\sqrt{-\dot{x}^2} + m\sqrt{-\dot{x}^2} = 0. \tag{11.16}$$

This is the case flagged at the end of §5.5. The matrix $\partial^2 L/\partial\dot{x}^\mu\partial\dot{x}^\nu$ annihilates $\dot{x}^\nu$, so the Legendre transform does not exist, and all the dynamics sits in the constraint $p^2 + m^2 = 0$. The reason is reparametrization invariance: there is no physical "s-time" for a Hamiltonian to generate.

Einbein action. From (11.6), $p_\mu = 2\dot{x}_\mu/(\alpha\lambda)$, and there is no momentum conjugate to λ. The Hamiltonian is

$$H = p_\mu \dot{x}^\mu - L = \lambda\,\frac{\alpha}{4}\big(p^2 + m^2\big), \qquad m = \frac{2\nu}{\alpha}. \tag{11.17}$$

The einbein multiplies the Hamiltonian, so it is a *Lagrange multiplier*: varying λ imposes

$$p^2 + m^2 = 0 \qquad \Longleftrightarrow \qquad E^2 = \mathbf{p}^2 + m^2, \qquad E = p^0. \tag{11.18}$$

This is the mass shell. Both signs $E = \pm\sqrt{\mathbf{p}^2 + m^2}$ satisfy it; the negative one returns in §14.4. The phase-space form of the action is

$$S[x, p, \lambda] = \int_0^1 \mathrm{d}s\,\Big[p_\mu \dot{x}^\mu - \lambda\,\frac{\alpha}{4}\big(p^2 + m^2\big)\Big], \tag{11.19}$$

the relativistic version of eq. (5.5). The momentum enters quadratically, so eliminating it is Case 1 of Part I (§5.3). With $p = 2\dot{x}/(\alpha\lambda)$,

$$p\cdot\dot{x} - \frac{\alpha\lambda}{4}\big(p^2 + m^2\big) = \frac{\dot{x}^2}{\alpha\lambda} - \frac{\alpha\lambda m^2}{4}, \tag{11.20}$$

which is the integrand of (11.6).

Hamilton–Jacobi. Replacing $p_\mu \to \partial_\mu S$ in the constraint gives the relativistic analogue of eq. (3.10), the Hamilton–Jacobi or eikonal equation:

$$\eta^{\mu\nu}\partial_\mu S\,\partial_\nu S + m^2 = 0. \tag{11.21}$$

It is solved by $S_{\mathrm{cl}}(x; x_i) = -m\sqrt{-(x - x_i)^2} = -m\tau$, since $\partial_\mu S_{\mathrm{cl}} = m(x - x_i)_\mu/\tau$ gives $(\partial S_{\mathrm{cl}})^2 = m^2(x - x_i)^2/\tau^2 = -m^2$. This is Hamilton's principal function of the free relativistic particle (§3.2). Dirac's proposal (§4.2) would give the amplitude $e^{-im\tau/\hbar}$; Chapter 13 finds what the full path integral adds to it.

12 Reparametrization as a gauge symmetry

12.1 Physical and gauge symmetries [hw. pp. 8–9]

The handwritten notes contrast two kinds of symmetry.

- **A physical symmetry.** The hydrogen atom has a spherically symmetric potential $V(r)$, so rotations $\mathbf{r} \to R(\theta)\mathbf{r}$ are a symmetry of the dynamics. A rotation still maps one state to a physically *different* one: rotating a $2p_z$ orbital gives a $2p_x$ orbital, which an experiment can tell apart. What the symmetry buys is degeneracy.
- **A gauge symmetry.** A reparametrization $s \to f(s)$ relabels the points of the same worldline. Nothing observable changes. The symmetry reflects a redundancy in the description, like the speed at which a pen traced a curve on paper: the ink is the same.

Configurations related by a gauge transformation are *physically equivalent*: they are one configuration described twice. For the path integral this means that summing over all pairs (x, λ) counts every physical configuration infinitely often, once for every element of the gauge group [hw. p. 9].

12.2 Dividing by the gauge volume [hw. pp. 8–10]

The cure is to divide by the volume V of the group of reparametrizations (11.10):

$$\langle x_f | x_i \rangle = \int \frac{\mathcal{D}x\,\mathcal{D}\lambda}{V}\, e^{iS[x,\lambda]/\hbar}. \tag{12.1}$$

The integrand is the same at every point of an orbit, so integrating along an orbit produces the factor V, which then cancels [hw. p. 10]. The gauge group acts on x and λ together, by (11.13).

> **Intuition: a finite-dimensional model**
>
> Integrate a rotation-invariant function $F(\mathbf{x}) = g(|\mathbf{x}|)$ over the plane:
>
> $$\int \mathrm{d}^2x\, F = \int_0^{2\pi} \mathrm{d}\theta \int_0^\infty \mathrm{d}r\, r\, g(r) = 2\pi \int_0^\infty \mathrm{d}r\, r\, g(r). \tag{12.2}$$
>
> The rotations play the role of the reparametrizations, the circles of radius r are the gauge orbits, and the half-line $\theta = 0$ is a gauge slice that meets each circle once. The factor 2π is the volume V of the group. The factor r is a Jacobian that remembers how big the orbit through each point of the slice is. It is the analogue of the "measure factor" $M(e)$ of [hw. p. 13], which §12.6 computes.

12.3 Orbits and slices [hw. pp. 10–11, 13]

Let $\mathcal{F}$ be the space of paths $x^\mu(s)$ and Σ the space of einbeins $\lambda(s)$, so that the path integral runs over $\mathcal{F} \times \Sigma$ [hw. p. 10]. The *gauge orbit* of λ is the set of all einbeins $\tilde{\lambda}$ reachable from λ by reparametrizations. A *gauge slice* is a subset of Σ that meets every orbit *exactly once* (Fig. 24). This is two conditions.

- **Uniqueness:** no two points of the slice lie on the same orbit, so nothing is counted twice. The handwritten notes state this condition [hw. p. 11].
- **Existence:** every orbit meets the slice, so nothing is lost. This condition is missing from the handwritten notes and is equally necessary.

The path integral is then restricted to the slice, and the integration along the orbits gives the factor V [hw. p. 12]. A slice is not unique. Applying a gauge transformation to a slice gives another slice, and any choice of one point from each orbit will do [hw. p. 11].

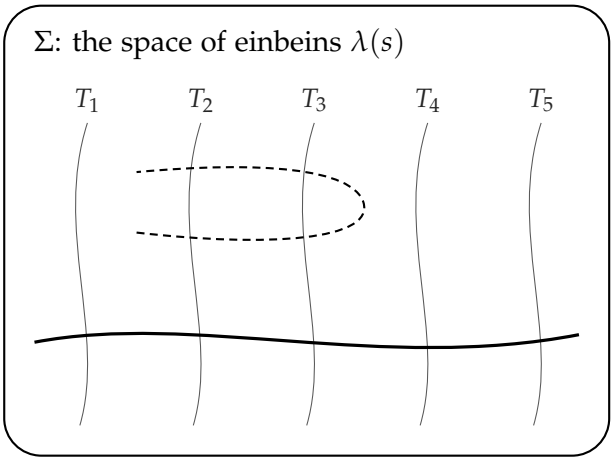


Figure 24: Gauge orbits (thin curves) in the space of einbeins. Each orbit is labelled by the one quantity a reparametrization cannot change, the proper-time length T (§12.4). A good slice, such as the constant einbeins, meets every orbit exactly once. A bad slice counts some configurations twice and misses others.

12.4 The proper-time length: the only invariant [hw. pp. 12–16]

Everything about the orbits follows from one observation. The total "length"

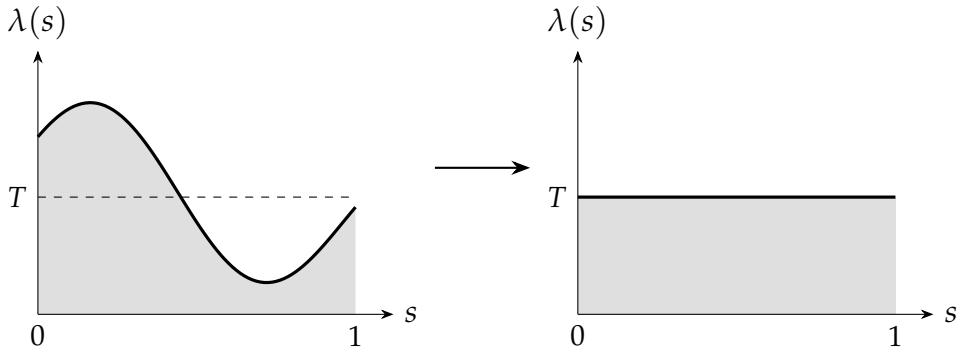
$$T[\lambda] \equiv \int_0^1 \lambda(s)\,\mathrm{d}s \tag{12.3}$$

is gauge invariant, because $\lambda\,\mathrm{d}s$ is invariant by (11.13): $\int_0^1 \tilde{\lambda}(\tilde{s})\,\mathrm{d}\tilde{s} = \int_0^1 (\lambda/\dot{f})\,\dot{f}\,\mathrm{d}s = T$.

- **Existence.** Every positive λ is gauge equivalent to the constant einbein $\lambda = T$. Take
$$f(s) = \frac{1}{T}\int_0^s \lambda(s')\,\mathrm{d}s'. \tag{12.4}$$
Then $f(0) = 0$, $f(1) = 1$, $\dot{f} = \lambda/T > 0$, and by (11.13) the new einbein is $\tilde{\lambda} = \lambda/\dot{f} = T$.

- **Uniqueness.** Two constants $T \neq T'$ are not gauge equivalent, because T is invariant. The handwritten argument reaches the same conclusion directly [hw. p. 12]: a reparametrization taking a constant to a constant has $\dot{f}$ constant, so $f(s) = c + ds$, and the endpoint conditions force $c = 0$, $d = 1$, the identity map $f(s) = s$.

So the constant einbeins form a good slice, and the orbits are labelled by a single positive number, the *modulus* $T \in (0, \infty)$. On the classical path $\lambda = \sqrt{-\dot{x}^2}/\nu$ (11.7), so $T = \tau_{\text{total}}/\nu$: the modulus is the proper time of the path in units of $1/\nu$. The gauge-fixed path integral sums over all paths and all proper durations.



a general einbein: area $T = \int_0^1 \lambda\,\mathrm{d}s$ — after the reparametrization (12.4)

Figure 25: A reparametrization redistributes the area under $\lambda(s)$ but cannot change it. The area T is the only gauge-invariant information in the einbein, and every positive einbein can be flattened to the constant T.

The sign of the einbein. The construction needs $\lambda > 0$ throughout, since $\dot{f} = \lambda/T$ must be positive. (The handwritten notes write $e \in \mathbb{R}$ on p. 13; it should be $e > 0$.) An einbein that is negative everywhere gives $T < 0$, a worldline traversed in the opposite orientation, and the sign of T is gauge invariant under orientation-preserving reparametrizations. We keep $T > 0$. The other choices produce other Green's functions (§13.6).

Caution: what the Fourier route of pp. 13–16 does, and two corrections

The handwritten notes prove existence by writing $\lambda = X + \delta\lambda$, with a fixed function X carrying prescribed endpoint values $\lambda(0) = a$, $\lambda(1) = b$, and expanding $\delta\lambda$ and δf in $\{1, \cos 2\pi ns, \sin 2\pi ns\}$. Solving $\dot{f} = \lambda/e$ mode by mode gives $f_n = -\theta_n/(2n\pi e)$, $g_n = \lambda_n/(2n\pi e)$ and $e = \int_0^1 X\,\mathrm{d}s + \lambda_0 = \int_0^1 \lambda\,\mathrm{d}s$: the modulus (12.3), reached the long way. Two points need correcting.

- The endpoint values of λ are not data. They change under reparametrizations, $\tilde{\lambda}(0) = \lambda(0)/\dot{f}(0)$, so they cannot be held fixed. Only x has boundary conditions.
- Every function in the basis takes the same value at $s = 0$ and $s = 1$. A convergent series therefore always has $X(0) = X(1)$, whatever the growth of its coefficients [hw. p. 15]. A non-periodic function is represented only in the mean-square sense: the coefficients fall like $1/n$, there is Gibbs overshoot near the ends, and the series gives $(a + b)/2$ at them.

The half-range bases avoid both problems: $\cos n\pi s$ for λ and $\sin n\pi s$ for the gauge parameter. They are used in §12.6. (Two typos on p. 16: the upper limit of the integral defining $Y(s)$ should be s, and the modulus integral runs over $(0, \infty)$, not $(0, 1)$.)

12.5 Why not fix the gauge on x? [hw. p. 10]

The handwritten notes observe that a reparametrization can bring λ to a simple form, and suggest that the functional form of $x^\mu(s)$ cannot be restricted in the same way. It can. The *static gauge*

$$x^0(s) = t_i + s\,(t_f - t_i) \tag{12.5}$$

is reachable for any path whose time coordinate increases monotonically: take f proportional to $x^0 - t_i$. The real objection is that it is *not global*. A path that turns back in time (Fig. 22b) cannot be brought to static gauge, so this gauge silently drops exactly the paths that describe antiparticles. In static gauge one is left with ordinary Schrödinger evolution generated by $\sqrt{\hat{\mathbf{p}}^2 + m^2}$: positive energies only, non-local in space, and not Lorentz covariant. The gauge $\lambda = T$ excludes nothing, because every positive einbein can be flattened (§12.4).

12.6 The Faddeev–Popov measure★

The handwritten notes write, for any gauge-invariant functional $g[\lambda]$,

$$\int \mathcal{D}\lambda\; g[\lambda] = V \int_0^\infty \mathrm{d}T\; M(T)\, g[T] \qquad \text{[hw. p. 13]}, \tag{12.6}$$

and skip the calculation of $M(T)$. Here it is.

Modes. Expand an einbein near the slice in the half-range cosine basis. The even extension of λ to $[-1,1]$ is continuous, so this series has none of the endpoint problems of the caution above:

$$\lambda(s) = T + \sum_{n\geq 1} a_n \cos n\pi s, \qquad T = \int_0^1 \lambda \, \mathrm{d}s. \tag{12.7}$$

The constant mode is the modulus, because every $\cos n\pi s$ with $n \geq 1$ integrates to zero. An infinitesimal reparametrization has $\xi(0) = \xi(1) = 0$, so it expands in the sine modes of Part I's oscillator (Fig. 19), $\xi(s) = \sum_{n\geq 1} \xi_n \sin n\pi s$. At the slice $\lambda = T$, eq. (11.14) gives

$$\delta\lambda = -T\dot{\xi} = -\sum_{n\geq 1} n\pi T \, \xi_n \cos n\pi s. \tag{12.8}$$

So the gauge directions are exactly the cosine modes with $n \geq 1$, with $a_n \leftrightarrow -n\pi T\xi_n$. The modulus is the zero mode and everything else is gauge.

The Jacobian. Changing variables from the a_n to the gauge parameters ξ_n,

$$\mathcal{D}\lambda = \mathrm{d}T \prod_{n\geq 1} \mathrm{d}a_n = \mathrm{d}T \; J(T) \prod_{n\geq 1} \mathrm{d}\xi_n, \qquad J(T) = \prod_{n\geq 1} n\pi T = \det\Big(T\frac{\mathrm{d}}{\mathrm{d}s}\Big)_{\text{Dirichlet}}. \tag{12.9}$$

The remaining integral $\int \prod \mathrm{d}\xi_n$ is the gauge volume V, which cancels. So $M(T) = J(T)$ is the *Faddeev–Popov determinant*: the Jacobian from (point on the slice, gauge parameters) to configurations. In the plane it was r.

What $J(T)$ is. It is an infinite product, and its T-dependence, $\prod_{n\geq 1} T$, has the same status as the constant C of §9. It must be defined together with the normalization of the x-measure, which also carries T-dependent factors (the $N(T)$ of §9.4). Different regularizations move powers of T between the two; only the product is meaningful. Physics fixes it. Suppose the net weight left over beyond the free-particle Gaussian were T^γ. In momentum space (§13.5) the propagator would become

$$\int_0^\infty \mathrm{d}T \; T^\gamma \, e^{-iaT(k^2+\mu^2-i\eta)} \;\propto\; \frac{1}{(k^2+\mu^2-i\eta)^{1+\gamma}}. \tag{12.10}$$

For $\gamma \neq 0$ this is not a Green's function of the Klein–Gordon operator. It is a Green's function of $(-\partial^2+\mu^2)^{1+\gamma}$ for integer γ, and of no differential operator otherwise. So the consistent measure is $M(T) = \text{const}$, normalized to 1 as on [hw. p. 16]. Section 14.2 confirms this independently: with $M = 1$ the amplitude is exactly the Green's function of the constraint.

Closed worldlines are different. On a circle the reparametrizations include the rigid rotations $s \to s + c$, which leave the constant einbein unchanged. The gauge group no longer acts freely, and these rotations must be divided out separately. On an interval the existence argument of §12.4 pinned $c = 0$ through $f(0) = 0$; on a circle nothing does. The result is an extra factor $1/T$ in the measure for closed loops, derived in §17.7.

13 The free relativistic propagator

13.1 The gauge-fixed path integral [hw. pp. 16–17]

By §12.4–§12.6, dividing by the gauge volume leaves an integral over the constant einbeins $\lambda = T$ with measure $\mathrm{d}T$:

$$\langle x_f | x_i \rangle = \int_0^\infty \mathrm{d}T \int_{x(0)=x_i}^{x(1)=x_f} \mathcal{D}x \; e^{iS_T[x]/\hbar}, \qquad S_T[x] = \frac{1}{\alpha}\int_0^1 \mathrm{d}s \left[\frac{\dot{x}^2}{T} - T\nu^2\right]. \tag{13.1}$$

This is the handwritten "working formula" of p. 17.

13.2 Proper-time gauge [hw. p. 17]

Rescale the parameter, $s' = Ts$ and $x'(s') = x(s)$. Then $T^{-1}\dot{x}^2\,\mathrm{d}s = (\mathrm{d}x'/\mathrm{d}s')^2 \mathrm{d}s'$ and $T\nu^2\,\mathrm{d}s = \nu^2\,\mathrm{d}s'$, so

$$S_T[x] = \frac{1}{\alpha}\int_0^T \mathrm{d}s' \left[\left(\frac{\mathrm{d}x'}{\mathrm{d}s'}\right)^2 - \nu^2\right]. \tag{13.2}$$

The handwritten notes say the T-dependence has been eliminated. It has moved: from the integrand into the length of the parameter interval. Dropping the primes,

$$\langle x_f | x_i \rangle = \int_0^\infty \mathrm{d}T \int_{x(0)=x_i}^{x(T)=x_f} \mathcal{D}x \; \exp\left\{\frac{i}{\alpha\hbar}\int_0^T \mathrm{d}s \left(\dot{x}^2 - \nu^2\right)\right\}. \tag{13.3}$$

At fixed T this is an ordinary quantum-mechanical propagator over a "time" T, for a free particle in D dimensions with Lagrangian $\dot{x}^2/\alpha$ (a particle of mass $2/\alpha$) in a constant potential ν^2/α. Everything in Part I applies to it.

13.3 Evaluating the Gaussian [hw. pp. 17–18]

Split $x = x_{\mathrm{cl}} + y$ with the straight line $x_{\mathrm{cl}}(s) = x_i + (x_f - x_i)\,s/T$ and $y(0) = y(T) = 0$. As in §9, the cross term vanishes because $\ddot{x}_{\mathrm{cl}} = 0$, so

$$\frac{1}{\alpha}\int_0^T \left(\dot{x}^2 - \nu^2\right)\mathrm{d}s = \frac{1}{\alpha}\left[\frac{(x_f - x_i)^2}{T} - T\nu^2\right] + \frac{1}{\alpha}\int_0^T \dot{y}^2\,\mathrm{d}s \tag{13.4}$$

and

$$\langle x_f | x_i \rangle = \int_0^\infty \mathrm{d}T \; \exp\left\{\frac{i}{\alpha\hbar}\left[\frac{(x_f - x_i)^2}{T} - T\nu^2\right]\right\} I(T), \tag{13.5}$$

with the fluctuation integral

$$I(T) = \int \mathcal{D}y \; \exp\left\{\frac{i}{\alpha\hbar}\int_0^T \mathrm{d}s\, \eta_{\mu\nu}\dot{y}^\mu\dot{y}^\nu\right\}. \tag{13.6}$$

Here $(x_f - x_i)^2 = \eta_{\mu\nu}(x_f - x_i)^\mu (x_f - x_i)^\nu$ carries the minus sign of the time component. The fluctuation integral factorizes, $I = \prod_{\mu=0}^{N} I_\mu$.

- **Space components.** Each is the free-particle normalization of eq. (9.11), with mass $2/\alpha$, time T and $x_f = x_i$: $I_j = \big(i\alpha\pi\hbar T\big)^{-1/2}$ [hw. p. 18].
- **The time component.** Its kinetic term has the opposite sign, $-\frac{i}{\alpha\hbar}\int(\dot{y}^0)^2$. Every Fresnel integral in its time slicing therefore has $a < 0$ in eq. (2.14), and gives $e^{-i\pi/4}$ instead of $e^{+i\pi/4}$. The measure is fixed by the mode's own kinetic term (§9), so $I_0 = \big(-i\alpha\pi\hbar T\big)^{-1/2}$, the complex-conjugate phase.

Since $(-i)^{-1/2}/(i)^{-1/2} = e^{i\pi/4}/e^{-i\pi/4} = i$,

$$I(T) = \big(i\alpha\pi\hbar T\big)^{-(D-1)/2}\big(-i\alpha\pi\hbar T\big)^{-1/2} = i\,\big(i\alpha\pi\hbar T\big)^{-D/2}. \tag{13.7}$$

The free relativistic propagator

$$K(x_f, x_i) \equiv \langle x_f | x_i\rangle = i\int_0^\infty \mathrm{d}T\,\big(i\alpha\pi\hbar T\big)^{-D/2}\exp\Big\{\frac{i}{\alpha\hbar}\Big[\frac{(x_f-x_i)^2}{T} - T\big(\nu^2 - i\eta\big)\Big]\Big\}. \tag{13.8}$$

Caution: the factor i and the second Wick rotation

The handwritten result on p. 18 lacks the overall i, because it uses the spatial factor $(i\alpha\pi\hbar T)^{-1/2}$ for the time component too. The second Wick rotation suggested there, $y^0 = -i\bar{y}^0$, is not needed: treat the time mode with its own sign. The factor is not cosmetic. With it the two comparisons with field theory in Chapter 16 agree exactly. §14.2 derives the same factor a second way, from operators.

13.4 Convergence and the $i\eta$ prescription★

At large T the factor $e^{-iT\nu^2/\alpha\hbar}$ oscillates without decaying, and the integral does not converge. The shift $\nu^2 \to \nu^2 - i\eta$ in (13.8) supplies the damping $e^{-\eta T/\alpha\hbar}$. Since $m = 2\nu/\alpha$, it is the same as

$$m^2 \to m^2 - i\eta, \tag{13.9}$$

the Feynman prescription in mostly-plus signature. In §8.9 the same kind of shift, $m\omega^2 \to m\omega^2 - i\eta$, was shown to be an infinitesimal clockwise rotation of time. Here it is an infinitesimal clockwise rotation of the proper-time length, $T \to T(1 - i\eta')$.

At small T the factor $e^{i(x_f-x_i)^2/\alpha\hbar T}$ oscillates infinitely fast. The integral converges there only as an oscillatory integral, just as the short-time free propagator of Part I tends to $\delta(x_f - x_i)$. In Euclidean form (§13.7) both ends converge absolutely.

13.5 Momentum space and Schwinger's representation★

The fixed-T kernel is a Gaussian in $x_f - x_i$, so it is a superposition of plane waves. For one space component, $\int \frac{\mathrm{d}k}{2\pi} e^{ik\Delta x - iak^2} = (4\pi ia)^{-1/2} e^{i\Delta x^2/4a}$. With $a = \alpha\hbar T/4$ this is $(i\alpha\pi\hbar T)^{-1/2} e^{i\Delta x^2/\alpha\hbar T}$. The time component has the opposite sign and gives the conjugate prefactor, so

$$i\left(i\alpha\pi\hbar T\right)^{-D/2} e^{i(x_f-x_i)^2/\alpha\hbar T} = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, e^{ik\cdot(x_f-x_i)}\, e^{-i(\alpha\hbar T/4)k^2}. \tag{13.10}$$

The mass term is $e^{-iTv^2/\alpha\hbar} = e^{-i(\alpha\hbar T/4)\mu^2}$ with

$$\mu \equiv \frac{2v}{\alpha\hbar} = \frac{m}{\hbar}, \tag{13.11}$$

the inverse (reduced) Compton wavelength. It is what the handwritten field-theory pages call "m" [hw. pp. 20–27]. Doing the T-integral first,

$$K(x_f,x_i) = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, e^{ik\cdot(x_f-x_i)} \int_0^\infty \mathrm{d}T\, e^{-i(\alpha\hbar T/4)(k^2+\mu^2-i\eta)} = \frac{4}{\alpha\hbar}\int \frac{\mathrm{d}^D k}{(2\pi)^D}\, \frac{-i\, e^{ik\cdot(x_f-x_i)}}{k^2+\mu^2-i\eta}. \tag{13.12}$$

Read backwards, this is *Schwinger's parametrization* of a propagator,

$$\frac{-i}{A-i\eta} = \int_0^\infty \mathrm{d}\sigma\, e^{-i\sigma(A-i\eta)} \qquad (A \text{ real}), \tag{13.13}$$

with $\sigma = \alpha\hbar T/4$. The modulus of the worldline *is* Schwinger's proper-time parameter. This is the remark of [hw. p. 28] that field-theory calculations in Schwinger's parametrization come out in worldline form. In the k^0-plane the poles of (13.12) sit at $k^0 = \pm(\omega_{\mathbf{k}} - i\eta')$, with $\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2+\mu^2}$. The positive-frequency pole lies below the real axis and the negative-frequency pole above it: the Feynman placement (Fig. 29).

13.6 Which Green's function? The range of T★

The range $T \in (0,\infty)$ was chosen, not derived: it came from $\lambda > 0$ (§12.4). Other ranges give other functions, as (13.13) shows with $A = k^2+\mu^2$:

range of $\sigma \propto T$	$\int \mathrm{d}\sigma\, e^{-i\sigma A}$ (regulated)	what it is
$(0,\infty)$	$\dfrac{-i}{k^2+\mu^2-i\eta}$	Feynman (time-ordered), §15.7
$(-\infty,0)$	$\dfrac{+i}{k^2+\mu^2+i\eta}$	anti-Feynman (anti-time-ordered)
$(-\infty,\infty)$	$2\pi\,\delta(k^2+\mu^2)$	on-shell, no source (Hadamard function)

The last line solves the *homogeneous* Klein–Gordon equation, because $(k^2+\mu^2)\delta(k^2+\mu^2)=0$. It is a solution, not a Green's function. The retarded function is missing from the table, and no range of T produces it. It needs both poles below the real k^0-axis, $k^0 \to k^0 + i\eta$, a prescription that depends on the sign of k^0 and not on k^2 alone. It is $\theta(x^0-y^0)$ times a commutator (§15.7).

> **Intuition: positive proper time is not forward time**
>
> In Part I a factor $\theta(T)$ in ordinary time, together with $+i\eta$, produced the retarded propagator, eq. (6.7). Here $\theta(T)$ in *proper* time produces the Feynman propagator instead. A worldline that runs backward in x^0 (Fig. 22b) still has a positive proper length. That is exactly the antiparticle (§14.4).

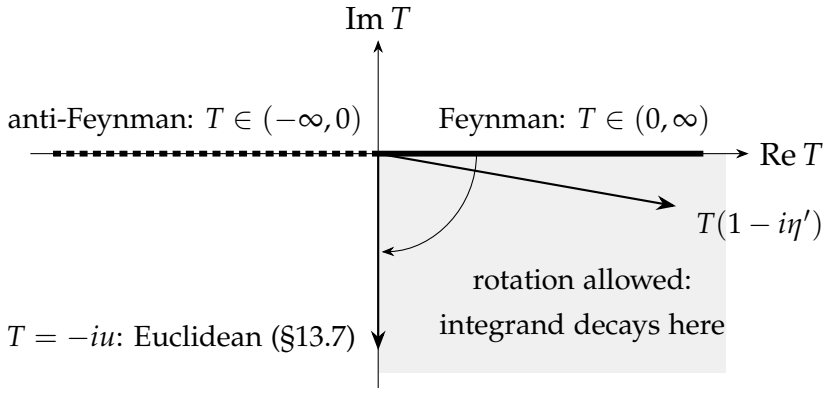


Figure 26: The complex plane of the proper-time length T. The Feynman propagator integrates over the positive real axis and the anti-Feynman propagator over the negative one; integrating over both gives an on-shell solution with no source. The $i\eta$ prescription tilts the contour infinitesimally into the shaded quadrant, where the integrand decays, and the Wick rotation turns it all the way to the negative imaginary axis. Compare Fig. 14.

13.7 The Euclidean propagator: Brownian motion in spacetime★

Continue the endpoints to Euclidean time, $x^0 = -i\bar{x}^0$, as on [hw. p. 25]. Then $(x_f - x_i)^2 \to r^2 \equiv |\bar{x}_f - \bar{x}_i|^2$, the positive Euclidean distance squared. Rotate the proper-time length as well, $T=-iu$ ("Wick rotate also the worldline time", [hw. p. 25]). The rotation is allowed because, for $\arg T \in [-\pi/2, 0]$, the factor $e^{-iT(v^2-i\eta)/\alpha\hbar}$ decays at large $|T|$ and $e^{ir^2/\alpha\hbar T}$ stays bounded (Fig. 26). With $\mathrm{d}T = -i\,\mathrm{d}u$, the factor $-i$ cancels the i of (13.8):

$$\begin{aligned}\bar{K}(r) &= \int_0^\infty \mathrm{d}u\,\left(\alpha\pi\hbar u\right)^{-D/2} e^{-r^2/\alpha\hbar u - uv^2/\alpha\hbar} \\ &= \frac{4}{\alpha\hbar}\int_0^\infty \mathrm{d}\sigma\,\left(4\pi\sigma\right)^{-D/2} e^{-r^2/4\sigma}\, e^{-\mu^2\sigma},\end{aligned} \tag{13.14}$$

with $\sigma = \alpha\hbar u/4$. Without the i of §13.3 the Euclidean propagator would come out as $-i$ times a positive function, as on [hw. p. 25].

Intuition: a relativistic random walk

The factor $(4\pi\sigma)^{-D/2}e^{-r^2/4\sigma}$ is the heat kernel of eq. (8.12), in D dimensions and with unit diffusion constant. It is the probability density for a Brownian particle to get from $\bar{x}_i$ to $\bar{x}_f$ in a "time" σ (§8.7). The factor $e^{-\mu^2\sigma}$ is the absorption of §8.7 with a constant potential: long excursions are suppressed. The Euclidean relativistic propagator is a sum over Brownian paths in *spacetime*, of every duration, each weighted by $e^{-\mu^2\sigma}$. It is the polymer of Fig. 17 with a price per unit length.

The integral is a standard Bessel representation:

$$\bar{K}(r) = \frac{4}{\alpha\hbar}\,\frac{1}{(2\pi)^{D/2}}\Big(\frac{\mu}{r}\Big)^{D/2-1}K_{D/2-1}(\mu r). \tag{13.15}$$

In $D = 4$ it is $\frac{4}{\alpha\hbar}\frac{\mu K_1(\mu r)}{4\pi^2 r}$, which tends to the massless $\frac{4}{\alpha\hbar}\frac{1}{4\pi^2 r^2}$ as $\mu \to 0$. In $D = 3$ it is $\frac{4}{\alpha\hbar}\frac{e^{-\mu r}}{4\pi r}$, the Yukawa potential. At short distance, $r \ll 1/\mu$, the mass is invisible and the propagator falls like $1/r^{D-2}$. At long distance, $r \gg 1/\mu$, it dies like $e^{-\mu r}$: *the range of the propagator is the Compton wavelength* (Fig. 27). This is why the exchange of a massive particle produces a short-range force. In momentum space (13.14) is $\frac{4}{\alpha\hbar}\int\frac{\mathrm{d}^D k}{(2\pi)^D}\frac{e^{ik\cdot(\bar{x}_f-\bar{x}_i)}}{k^2+\mu^2}$ with a Euclidean k, positive like a probability (§15.5).

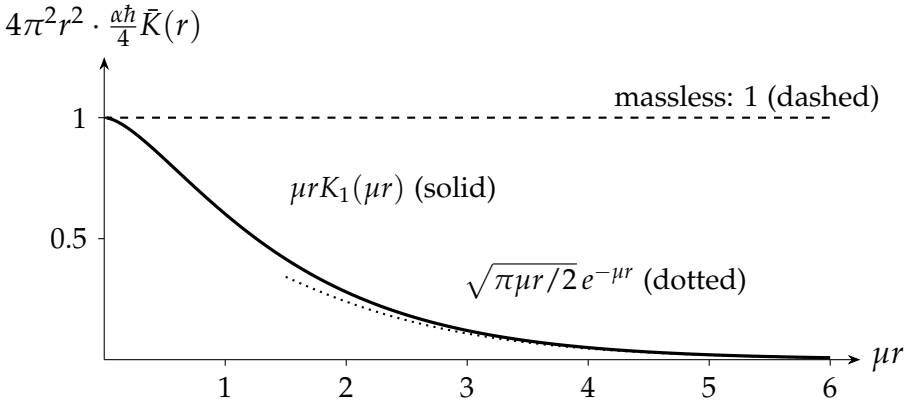


Figure 27: The Euclidean propagator in $D = 4$, multiplied by $4\pi^2r^2$ so that the massless propagator becomes the constant 1. Below a Compton wavelength ($\mu r \lesssim 1$) the mass does not matter; beyond it the propagator dies exponentially, following the dotted asymptote. The same curve describes the overlap of one-particle states at equal times (§15.6).

13.8 Two checks: the classical and the non-relativistic limits★

The classical limit. Apply stationary phase (§2.5) to the T-integral of (13.8). Its phase is $\varphi(T) = \frac{1}{\alpha\hbar}\left[\Delta x^2/T - Tv^2\right]$. For a timelike separation, $\Delta x^2 = -\tau^2$, and

$$\varphi'(T) = \frac{1}{\alpha\hbar}\left[\frac{\tau^2}{T^2} - v^2\right] = 0 \implies T_* = \frac{\tau}{v}, \qquad \varphi(T_*) = -\frac{2v\tau}{\alpha\hbar} = -\frac{m\tau}{\hbar}. \tag{13.16}$$

So $K \sim e^{-im\tau/\hbar} = e^{iS_{\rm cl}/\hbar}$, with $S_{\rm cl} = -m\tau$ the Hamilton principal function of §11.6. The square-root action reappears, this time quantum mechanically, and the saddle $T_* = \tau/v$ is the on-shell modulus of §12.4. Doing the T-integral by stationary phase is the quantum version of eliminating the einbein (§11.4).

For a spacelike separation, $\Delta x^2 = r^2 > 0$, the saddle moves to imaginary $T_* = -ir/v$ (on the rotated contour of Fig. 26), and $i\varphi(T_*) = -2vr/\alpha\hbar = -\mu r$. The amplitude leaks outside the light cone, but only over a Compton wavelength (Fig. 28). This does not violate causality (§15.7).

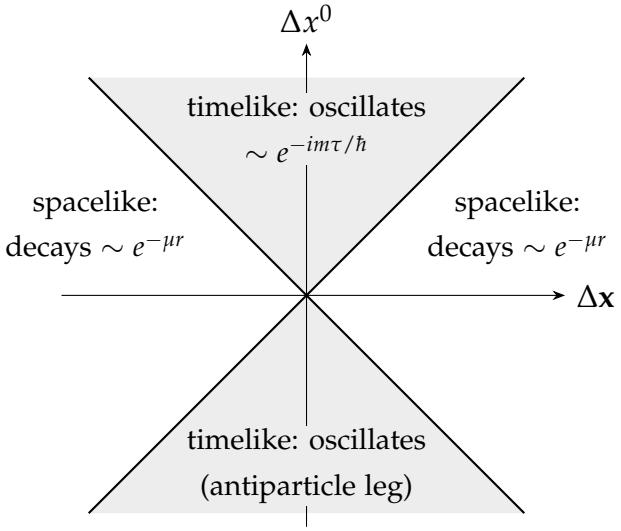


Figure 28: Where the Feynman propagator lives. Inside the light cone (shaded) it oscillates with the phase $e^{-im\tau/\hbar}$ of the classical action. Outside it is not zero but decays over a Compton wavelength, an evanescent tail like tunnelling under a barrier.

The non-relativistic limit. Take $t \equiv x_f^0 - x_i^0 > 0$ and do the k^0-integral of (13.12) by residues. Close the contour below and pick up the pole $k^0 = \omega_{\mathbf{k}}$:

$$K(x_f, x_i) = \frac{4}{\alpha\hbar}\int \frac{\mathrm{d}^N k}{(2\pi)^N}\,\frac{e^{i\mathbf{k}\cdot\Delta\mathbf{x} - i\omega_{\mathbf{k}}t}}{2\omega_{\mathbf{k}}} \qquad (t>0). \tag{13.17}$$

When the momenta that matter are small, $|\mathbf{k}| \ll \mu$, use $\omega_{\mathbf{k}} \approx \mu + \mathbf{k}^2/2\mu$ and $1/2\omega_{\mathbf{k}} \approx 1/2\mu$. The remaining integral is a Gaussian:

$$K \approx \frac{2}{\alpha\hbar\mu}\,e^{-i\mu t}\int \frac{\mathrm{d}^N k}{(2\pi)^N}\,e^{i\mathbf{k}\cdot\Delta\mathbf{x} - i\mathbf{k}^2 t/2\mu} = \frac{2}{\alpha m}\,e^{-imc^2 t/\hbar}\,K_{\rm free}(\mathbf{x}_f, t; \mathbf{x}_i, 0), \tag{13.18}$$

where $K_{\text{free}} = \left(\frac{m}{2\pi i\hbar t}\right)^{N/2} e^{im|\Delta\mathbf{x}|^2/2\hbar t}$ is Part I's free propagator, eq. (9.11), in N dimensions. The approximation needs $|\Delta\mathbf{x}| \ll t$ and $t \gg \hbar/mc^2$. The result carries three lessons.

- The phase $e^{-imc^2t/\hbar}$ is the rest energy. Non-relativistic quantum mechanics measures energies from mc^2, which it may, since shifting $\hat{H}$ by a constant changes nothing (footnote in §8.3).
- The factor $1/2\omega_{\mathbf{k}} \to 1/2\mu$ is the relativistic normalization of states (§15.6).
- For $t < 0$ the other pole contributes, with the negative-frequency factor $e^{+i\omega_{\mathbf{k}}t}$. That is the antiparticle, which the non-relativistic theory leaves out.

14 The propagator as a Green's function, again

14.1 The question [hw. pp. 22–23]

In Part I the propagator turned out to be a Green's function. The retarded propagator $K_r = \theta(t - t_i)\langle x, t|x_i, t_i\rangle$ satisfies, eq. (6.5),

$$\big[i\hbar\partial_t - H\big]K_r(x, t; x_i, t_i) = i\hbar\,\delta(t - t_i)\,\delta(x - x_i). \tag{14.1}$$

(The handwritten line on p. 22 writes K_r as $\langle x, t|x_i, t_i\rangle$, without the step function.) The handwritten notes ask whether the worldline amplitude $K(x_f, x_i)$ is also a Green's function, and of which equation [hw. p. 23]. They answer through field theory (Chapters 15–16). The worldline theory can answer it on its own, and that is the subject of this chapter.

14.2 The worldline Hamiltonian and its resolvent★

Quantize the phase-space action (11.19). The coordinates $\hat{x}^\mu$ and momenta $\hat{p}_\mu$ satisfy $[\hat{x}^\mu, \hat{p}_\nu] = i\hbar\,\delta^\mu_\nu$ and act on wavefunctions $\psi(x)$ of all D coordinates. This is an *enlarged* Hilbert space, in which even $\hat{x}^0$ is an operator, unlike in Part I. Its position eigenstates satisfy $\langle x|p\rangle = e^{ip\cdot x/\hbar}/(2\pi\hbar)^{D/2}$. In the gauge $\lambda = T$, the action (11.19) has Hamiltonian $T\hat{H}$ acting for unit parameter time, with

$$\hat{H} \equiv \frac{\alpha}{4}(\hat{p}^2 + m^2). \tag{14.2}$$

At fixed T the gauge-fixed path integral is therefore the phase-space path integral of Part I, eq. (5.4), for this Hamiltonian. The Hamiltonian is quadratic in the momenta (Case 1, §5.3), so the path integral equals the matrix element $\langle x_f|e^{-iT\hat{H}/\hbar}|x_i\rangle$.

A check of the factor i. Compute the matrix element directly, inserting momentum eigenstates:

$$\langle x_f|e^{-iT\hat{H}/\hbar}|x_i\rangle = \int \frac{\mathrm{d}^D p}{(2\pi\hbar)^D}\; e^{ip\cdot(x_f - x_i)/\hbar}\, e^{-i(\alpha T/4\hbar)(-p_0^2 + \mathbf{p}^2 + m^2)}. \tag{14.3}$$

Each space component is a Fresnel integral with $a < 0$ in eq. (2.14). The time component has $a > 0$ and contributes the conjugate phase. Together they reproduce exactly the integrand of (13.8), including the overall i of §13.3.

The resolvent. Now integrate over T, with the regulator of §13.4:

$$K(x_f, x_i) = \int_0^\infty \mathrm{d}T\,\langle x_f|e^{-iT(\hat{H} - i\eta)/\hbar}|x_i\rangle = \left\langle x_f\middle|\frac{-i\hbar}{\hat{H} - i\eta}\middle|x_i\right\rangle. \tag{14.4}$$

The relativistic propagator is Part I's retarded Green's function at zero energy

$$K(x_f, x_i) = \left\langle x_f \right| \frac{i\hbar}{E - \hat{H} + i\eta} \left| x_i \right\rangle \Big|_{E=0}, \qquad \hat{H} = \frac{\alpha}{4}(\hat{p}^2 + m^2), \tag{14.5}$$

which is $\hat{G}_r(E)$ of eq. (6.7) evaluated at $E = 0$.

Applying $\hat{H}$ to (14.4) gives $\hat{H}K = -i\hbar\langle x_f|x_i\rangle = -i\hbar\,\delta^D(x_f - x_i)$. In the position representation $\hat{p}_\mu = -i\hbar\partial_\mu$ and $\hat{p}^2 = -\hbar^2\partial^2$, with $\partial^2 = \eta^{\mu\nu}\partial_\mu\partial_\nu = -\partial_t^2 + \nabla^2$. Hence

$$\begin{gathered} \frac{\alpha}{4}\big(-\hbar^2\partial^2 + m^2\big)K(x, x_i) = -i\hbar\,\delta^D(x - x_i), \quad \text{that is,} \\ \big(-\partial^2 + \mu^2\big)K(x, x_i) = -\frac{4i}{\alpha\hbar}\,\delta^D(x - x_i). \end{gathered} \tag{14.6}$$

The operator $-\partial^2 + \mu^2 = \partial_t^2 - \nabla^2 + \mu^2$ is the *Klein–Gordon operator*. So the worldline amplitude is a Green's function of the Klein–Gordon equation, and the analysis of §13.6 says which one: the Feynman Green's function. This answers the question of [hw. p. 23] without any field theory.

> **Intuition: why zero energy?**
>
> In Part I the energy representation came from Fourier transforming in time, $\int \mathrm{d}T\, e^{iET/\hbar}\theta(T)e^{-i\hat{H}T/\hbar}$. Here there is no physical energy to transform to. The Hamiltonian is a constraint, and physical states must satisfy $\hat{H}\psi = 0$ (§14.3). The integral over proper durations is the same transform, evaluated at the only value the constraint allows. Physical states sit at $E = 0$, and the propagator is the resolvent there.

> **Caution: what $\langle x_f|x_i\rangle$ means**
>
> In the enlarged Hilbert space $\langle x_f|x_i\rangle$ is literally $\delta^D(x_f - x_i)$. The amplitude that the handwritten notes write as $\langle x_f|x_i\rangle$ is really $\langle x_f|\hat{G}|x_i\rangle$, with $\hat{G} = \int_0^\infty \mathrm{d}T\, e^{-iT\hat{H}/\hbar}$, the sum over all proper durations. The states $|x\rangle$ are not physical states: they do not satisfy the constraint.

14.3 First quantization: the Klein–Gordon equation as a constraint★

Dirac's rule for a constrained system is to impose the constraint on the states. Here it gives

$$\hat{H}\psi = 0 \quad\Longleftrightarrow\quad \big(-\hbar^2\partial^2 + m^2\big)\psi = 0 \quad\Longleftrightarrow\quad \big(\partial_t^2 - \nabla^2 + \mu^2\big)\psi(x) = 0, \tag{14.7}$$

the Klein–Gordon equation. A physical state does not evolve in s; the worldline has no physical time. The coordinate time x^0 is simply one of the arguments of ψ, and all the dynamics is in the constraint. Its plane-wave solutions are $e^{ik\cdot x}$ with $k^2 + \mu^2 = 0$, that is, $k^0 = \pm\omega_{\mathbf{k}}$.

Remark 14.1 (Where this leads). The structure found here is a reparametrization-invariant action, a Hamiltonian that is a constraint, and physical states annihilated by it. It is exactly the structure of canonical quantum gravity, where the constraint is the Wheeler–DeWitt equation $\hat{H}\Psi = 0$ and the absence of an external time is the "problem of time". The relativistic particle is the simplest model of it. The choice of range for T, which decided between a Green's function and a solution in §13.6, reappears in quantum cosmology as the choice of contour for the lapse.

14.4 Negative frequencies and antiparticles★

The Klein–Gordon equation is second order in time, so it has both positive-frequency ($e^{-i\omega t}$) and negative-frequency ($e^{+i\omega t}$) solutions, and both are needed for a complete set. Its conserved inner product is

$$(\psi_1, \psi_2) = i \int \mathrm{d}^N x \, \left(\psi_1^* \, \partial_t \psi_2 - (\partial_t \psi_1^*) \, \psi_2\right). \tag{14.8}$$

For $\psi = e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}$ the density $i(\psi^* \partial_t \psi - \partial_t \psi^* \, \psi)$ is $2\omega > 0$, but for $e^{+i\omega t + i\mathbf{k}\cdot\mathbf{x}}$ it is $-2\omega < 0$. There is no positive probability density for a single relativistic particle.

The way out, due to Stückelberg and Feynman, is to read a negative-frequency solution running backward in x^0 as a positive-energy *antiparticle* running forward. In the worldline sum this is literal. A path that turns back in time (Fig. 22b) is, read forward in x^0, a pair created at its lower turning point, whose antiparticle later annihilates the incoming particle at the upper turning point. The Feynman propagator for $x^0 < y^0$ is this antiparticle leg (§15.7). The non-relativistic limit of §13.8, which kept only $t > 0$, dropped it.

Why one relativistic particle is not enough.

1. The inner product (14.8) is not positive, so there is no probability density.
2. A relativistic particle cannot be localized better than about a Compton wavelength $\hbar/mc$ (§15.6).
3. Localizing it more sharply needs momenta $\gtrsim mc$, hence energy uncertainties of order mc^2, enough to create pairs. The number of particles is not fixed.
4. The amplitude is nonzero outside the light cone (§13.8). This is harmless only once antiparticles are included, which makes the relevant commutator vanish (§15.7).

The consistent framework is a *field*, the subject of the next chapter. The worldline survives as the description of the field's propagator (Chapter 16).

15 The Klein–Gordon field

The previous chapter ended with the need for a field. This chapter collects the facts about the free Klein–Gordon field needed to compare it with the worldline (Chapter 16). The guiding idea comes from Part I's last chapter: a free field is a collection of harmonic oscillators.

15.1 Each mode is an oscillator [hw. pp. 19–21]

The Klein–Gordon action is

$$S_{\mathrm{KG}} = -\frac{1}{2}\int \mathrm{d}^D x\,\left(\partial_\mu\phi\,\partial^\mu\phi + \mu^2\phi^2\right) = \int \mathrm{d}t \int \mathrm{d}^N x\,\left[\tfrac{1}{2}\dot\phi^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}\mu^2\phi^2\right]. \quad (15.1)$$

Fourier transform in space, $\phi(t,\mathbf{x}) = \int \frac{\mathrm{d}^N k}{(2\pi)^{N/2}}\,\tilde\phi(t,\mathbf{k})\,e^{i\mathbf{k}\cdot\mathbf{x}}$. A real field has $\tilde\phi(t,-\mathbf{k}) = \tilde\phi(t,\mathbf{k})^*$. Then

$$S_{\mathrm{KG}} = \int \mathrm{d}t \int \mathrm{d}^N k\;\tfrac{1}{2}\Big[\,|\dot{\tilde\phi}(t,\mathbf{k})|^2 - \omega_{\mathbf{k}}^2\,|\tilde\phi(t,\mathbf{k})|^2\Big], \qquad \omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2+\mu^2}. \quad (15.2)$$

Each wave vector, paired with its opposite, is a harmonic oscillator of unit mass and frequency $\omega_{\mathbf{k}}$, the system of §9. The oscillator's "position" is the field amplitude $\tilde\phi(\mathbf{k})$, not a position in space; Chapter 18 returns to this point. The handwritten notes set out the analogy as a dictionary [hw. pp. 20–21]:

oscillator (mass 1)	Klein–Gordon field
$S = \frac{1}{2}\int \mathrm{d}t\,(\dot x^2 - \omega^2 x^2)$	$S = -\frac{1}{2}\int \mathrm{d}^D x\,(\partial\phi\cdot\partial\phi + \mu^2\phi^2)$
$\hat H = \frac{1}{2}\hat p^2 + \frac{1}{2}\omega^2\hat x^2 = \hbar\omega(\hat a^\dagger\hat a + \frac{1}{2})$	$\hat H = \int \mathrm{d}^N k\;\hbar\omega_{\mathbf{k}}\,\hat a^\dagger_{\mathbf{k}}\hat a_{\mathbf{k}}$ (normal-ordered)
$\hat x(t) = \sqrt{\frac{\hbar}{2\omega}}\,(\hat a e^{-i\omega t} + \hat a^\dagger e^{i\omega t})$	$\hat{\tilde\phi}(t,\mathbf{k}) = \sqrt{\frac{\hbar}{2\omega_{\mathbf{k}}}}\,(\hat a_{\mathbf{k}} e^{-i\omega_{\mathbf{k}} t} + \hat a^\dagger_{-\mathbf{k}} e^{i\omega_{\mathbf{k}} t})$
$\hat p(t) = -i\sqrt{\frac{\hbar\omega}{2}}\,(\hat a e^{-i\omega t} - \hat a^\dagger e^{i\omega t})$	$\hat{\tilde\Pi}(t,\mathbf{k}) = -i\sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2}}\,(\hat a_{\mathbf{k}} e^{-i\omega_{\mathbf{k}} t} - \hat a^\dagger_{-\mathbf{k}} e^{i\omega_{\mathbf{k}} t})$
$[\hat x,\hat p] = i\hbar,\ \ [\hat a,\hat a^\dagger] = 1$	$[\hat\phi(t,\mathbf{x}),\hat\Pi(t,\mathbf{y})] = i\hbar\,\delta^N(\mathbf{x}-\mathbf{y}),\ \ [\hat a_{\mathbf{k}},\hat a^\dagger_{\mathbf{q}}] = \delta^N(\mathbf{k}-\mathbf{q})$
$\Delta(t-t') = i\langle 0\vert\mathcal{T}\,\hat x(t)\hat x(t')\vert 0\rangle$	$\Delta(x-y) = i\langle 0\vert\mathcal{T}\,\hat\phi(x)\hat\phi(y)\vert 0\rangle$
$(\partial_t^2 + \omega^2)\Delta = \hbar\,\delta(t-t')$	$(-\partial^2 + \mu^2)\Delta = \hbar\,\delta^D(x-y)$

Caution: three corrections to the dictionary

- The oscillator Hamiltonian is $\hbar\omega(\hat a^\dagger\hat a + \frac{1}{2})$. The handwritten $\hbar\omega(\hat a\hat a^\dagger + \frac{1}{2})$ [hw. p. 20] equals $\hbar\omega(\hat a^\dagger\hat a + \frac{3}{2})$.
- The field Hamiltonian needs its $\hbar$, and it is normal-ordered. The zero-point energy $\frac{1}{2}\hbar\omega_{\mathbf{k}}$ per mode, kept for the oscillator, has been dropped. Kept, it would

be $\int \mathrm{d}^N k\, \frac{1}{2}\hbar\omega_{\mathbf{k}}\, \delta^N(0)$, infinite; Part III deals with it (§24.4).

- The Wightman function needs $\hbar$ (next section), to match the oscillator's $\frac{\hbar}{2\omega}e^{-i\omega\Delta t}$ [hw. p. 21].

15.2 Mode expansion [hw. pp. 20–21, 39]

In position space, $\hat{\phi}(x) = \int \frac{\mathrm{d}^N k}{(2\pi)^{N/2}}\hat{\tilde{\phi}}(t,\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}}$ and similarly for $\hat{\Pi} = \partial_t\hat{\phi}$. Renaming $\mathbf{k} \to -\mathbf{k}$ in the $\hat{a}^\dagger$ term combines the two exponentials into covariant ones:

$$\hat{\phi}(x) = \int \frac{\mathrm{d}^N k}{(2\pi)^{N/2}}\sqrt{\frac{\hbar}{2\omega_{\mathbf{k}}}}\Big(\hat{a}_{\mathbf{k}}\, e^{ik\cdot x} + \hat{a}^\dagger_{\mathbf{k}}\, e^{-ik\cdot x}\Big), \qquad k^0 = \omega_{\mathbf{k}}, \quad k\cdot x = -\omega_{\mathbf{k}}t + \mathbf{k}\cdot\mathbf{x}. \tag{15.3}$$

The equal-time commutator of the field becomes $[\hat{\tilde{\phi}}(t,\mathbf{k}), \hat{\tilde{\Pi}}(t,\mathbf{q})] = i\hbar\,\delta^N(\mathbf{k}+\mathbf{q})$ in momentum space (exercise of [hw. p. 39], solved in Appendix C).

15.3 Green's functions [hw. p. 21]

The *Wightman function* is

$$D(x-y) \equiv \langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle = \int \frac{\mathrm{d}^N k}{(2\pi)^N}\,\frac{\hbar}{2\omega_{\mathbf{k}}}\, e^{ik\cdot(x-y)}, \qquad k^0 = \omega_{\mathbf{k}}, \tag{15.4}$$

since only the term $\hat{a}_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{q}}$ survives between vacua. The *Feynman function* is the time-ordered product, and the handwritten notes define Δ as i times it:

$$\begin{gathered} G_{\mathrm{F}}(x-y) \equiv \langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle = \theta(x^0-y^0)D(x-y) + \theta(y^0-x^0)D(y-x), \\ \Delta \equiv iG_{\mathrm{F}}. \end{gathered} \tag{15.5}$$

It is a Green's function of the Klein–Gordon operator:

$$\big(-\partial^2 + \mu^2\big)\Delta(x-y) = \hbar\,\delta^D(x-y). \tag{15.6}$$

Mode by mode this is the oscillator identity $(\partial_t^2 + \omega_{\mathbf{k}}^2)\Delta_{\mathbf{k}} = \hbar\,\delta(t)$, which is Part I's contact term, eq. (7.11), with $m = 1$, evaluated in the vacuum (proof in Appendix C).

15.4 Energy and momentum representations [hw. p. 22]

For the oscillator,

$$\Delta(t) = -\hbar\int_{-\infty}^{\infty}\frac{\mathrm{d}E}{2\pi}\,\frac{e^{-iEt}}{E^2-\omega^2+i\eta}. \tag{15.7}$$

The poles sit at $E = \pm(\omega - i\eta')$, and closing the contour below for $t > 0$ and above for $t < 0$ gives $\Delta = \frac{i\hbar}{2\omega}e^{-i\omega|t|}$ (Appendix C). For the field take $E = k^0$, and note that

$E^2 - \omega_{\mathbf{k}}^2 + i\eta = -(k^2 + \mu^2 - i\eta)$ in mostly-plus signature:

$$\Delta(x-y) = \hbar \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \frac{e^{ik\cdot(x-y)}}{k^2 + \mu^2 - i\eta}. \tag{15.8}$$

Compare the worldline result (13.12): $K = -\frac{4i}{\alpha\hbar^2}\Delta$, with $\mu = m/\hbar$. The two are already proportional. Chapter 16 explains why.

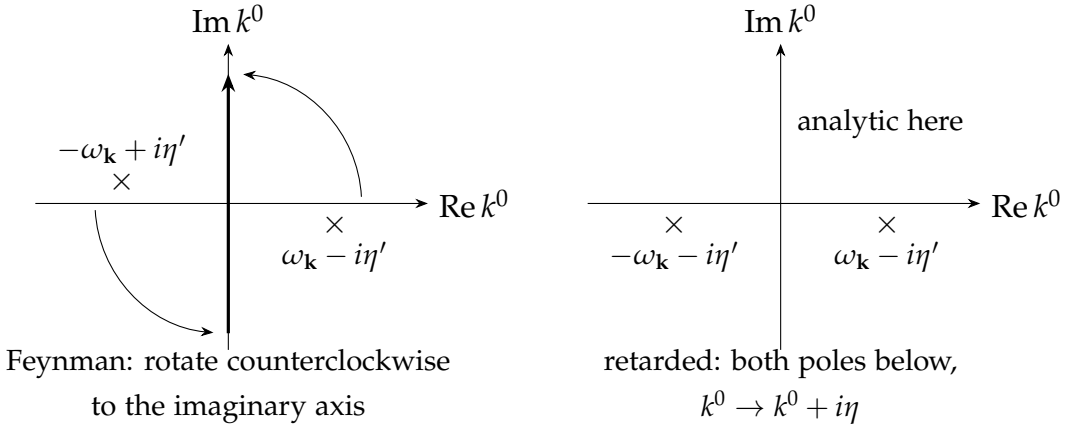


Figure 29: Poles in the complex k^0-plane. Left: the Feynman placement. The contour can be rotated counterclockwise onto the imaginary axis (the Wick rotation of §15.5) without crossing a pole, because the rotation sweeps the first and third quadrants. Right: the retarded placement, compare Fig. 10.

15.5 Euclidean version [hw. p. 22]

Define $\bar{\Delta}(\bar{x} - \bar{y}) \equiv \Delta(x-y)|_{x^0 = -i\bar{x}^0}$, with the branch of Part I. In the k^0-integral rotate the contour counterclockwise, $k^0 = i\bar{k}^0$ (Fig. 29): time rotates clockwise and energy counterclockwise, as in the remark of §8.3. Then $k \cdot x \to$ (after $\bar{k}^0 \to -\bar{k}^0$) the Euclidean product $\bar{k} \cdot \bar{x} = \bar{k}^0 \bar{x}^0 + \mathbf{k} \cdot \mathbf{x}$, $k^2 + \mu^2 \to \bar{k}^2 + \mu^2$, and $\mathrm{d}k^0 = i\,\mathrm{d}\bar{k}^0$:

$$\bar{\Delta}(\bar{x} - \bar{y}) = i\hbar \int \frac{\mathrm{d}^D \bar{k}}{(2\pi)^D} \, \frac{e^{i\bar{k}\cdot(\bar{x}-\bar{y})}}{\bar{k}^2 + \mu^2}. \tag{15.9}$$

The factor i comes from the measure. The Euclidean time-ordered function, $-i\bar{\Delta} = \hbar \int \frac{\mathrm{d}^D \bar{k}}{(2\pi)^D} \frac{e^{i\bar{k}\cdot(\bar{x}-\bar{y})}}{\bar{k}^2+\mu^2}$, is positive, like the probabilities of §8.7. Comparing with (13.14), $\bar{K} = \frac{4}{\alpha\hbar^2}\,(-i\bar{\Delta})$.

15.6 One-particle states and the limits of localization [hw. pp. 23–24]

The vacuum satisfies $\hat{a}_{\mathbf{k}}|0\rangle = 0$ for all $\mathbf{k}$. One-particle momentum states are normalized relativistically [hw. p. 23]:

$$|\mathbf{k}\rangle \equiv \sqrt{2\omega_{\mathbf{k}}}\, \hat{a}_{\mathbf{k}}^\dagger |0\rangle, \qquad \langle \mathbf{k}|\mathbf{q}\rangle = 2\omega_{\mathbf{k}}\, \delta^N(\mathbf{k} - \mathbf{q}). \tag{15.10}$$

This normalization is Lorentz invariant, because $\mathrm{d}^N k/2\omega_{\mathbf{k}}$ is the invariant measure on the mass shell, $\int \mathrm{d}^D k\, \delta(k^2+\mu^2)\theta(k^0) f(k) = \int \frac{\mathrm{d}^N k}{2\omega_{\mathbf{k}}} f(k)$ (aside of [hw. p. 23]; Appendix C). Acting on the vacuum, the field creates a superposition of them [hw. p. 24]:

$$\hat{\phi}(x)|0\rangle = \sqrt{\hbar}\int \frac{\mathrm{d}^N k}{(2\pi)^{N/2}} \frac{1}{2\omega_{\mathbf{k}}} e^{-ik\cdot x}|\mathbf{k}\rangle, \qquad \langle \mathbf{q}|\hat{\phi}(0,\mathbf{x})|0\rangle = \sqrt{\hbar}\,\frac{e^{-i\mathbf{q}\cdot\mathbf{x}}}{(2\pi)^{N/2}}. \tag{15.11}$$

The second formula is the relativistic analogue of $\langle p|x\rangle = e^{-ipx/\hbar}/\sqrt{2\pi\hbar}$. So $|0,\mathbf{x}\rangle \equiv \hat{\phi}(0,\mathbf{x})|0\rangle$ creates a particle "at $\mathbf{x}$", and at other times $|x\rangle \equiv \hat{\phi}(x)|0\rangle = e^{i\hat{H}t/\hbar}|0,\mathbf{x}\rangle$, using $\hat{H}|0\rangle = 0$ [hw. p. 24]. This is the same convention as Part I's Heisenberg states $|x,t\rangle = e^{i\hat{H}t/\hbar}|x\rangle$. There is no position operator in field theory: $\mathbf{x}$ is a label, like t.

> **Caution: these are not position eigenstates**
>
> States created at different points are not orthogonal:
>
> $$\langle 0,\mathbf{x}|0,\mathbf{y}\rangle = D(\mathbf{x}-\mathbf{y},0) = \hbar\int \frac{\mathrm{d}^N k}{(2\pi)^N}\frac{e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}}{2\omega_{\mathbf{k}}} \overset{N=3}{=} \hbar\,\frac{\mu K_1(\mu r)}{4\pi^2 r}, \qquad r = |\mathbf{x}-\mathbf{y}|. \tag{15.12}$$
>
> This is the same function as the Euclidean propagator in $D = 4$ (Fig. 27). It decays only like $e^{-\mu r}$, so a "particle at $\mathbf{x}$" is smeared over a Compton wavelength. The "unusual factor" $1/2\omega_{\mathbf{k}}$ noticed on [hw. p. 24] is responsible. With $1/\sqrt{2\omega_{\mathbf{k}}}$ instead (Newton–Wigner states), states at different points become orthogonal, but they no longer transform covariantly under boosts. A relativistic particle cannot be localized better than about $\hbar/mc$, as §14.4 anticipated.

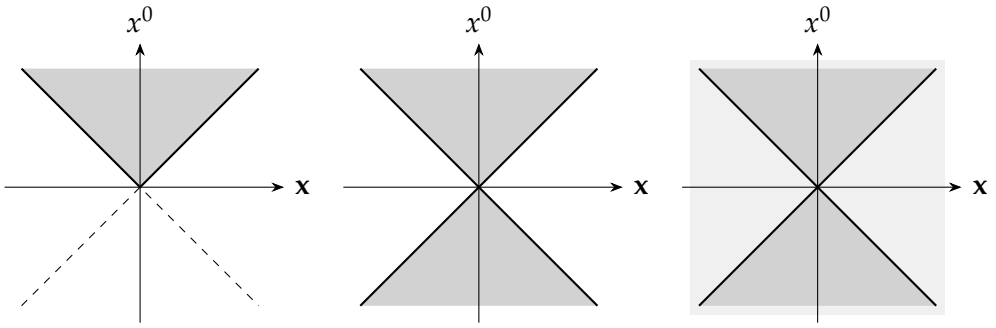
15.7 What "causal" means: Feynman versus retarded★

Four two-point functions appear, and the handwritten notes use the word "causal" for one of them [hw. pp. 22–25]. They are:

$$\begin{aligned}
&\text{Wightman:} && D(x-y) = \langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle,\\
&\text{commutator:} && C(x-y) = \langle 0|[\hat{\phi}(x),\hat{\phi}(y)]|0\rangle = D(x-y) - D(y-x),\\
&\text{retarded:} && \Delta_{\mathrm{R}}(x-y) = i\,\theta(x^0-y^0)\,C(x-y),\\
&\text{Feynman:} && \Delta(x-y) = i\big[\theta(x^0-y^0)D(x-y) + \theta(y^0-x^0)D(y-x)\big].
\end{aligned} \tag{15.13}$$

Both Δ and Δ_{R} satisfy $(-\partial^2+\mu^2)\Delta = \hbar\,\delta^D$ (Appendix C). They differ by boundary conditions, exactly as in §6.4: the equation does not fix its Green's function, and two Green's functions differ by a solution of the homogeneous equation.

Causality is a statement about the commutator. For a spacelike separation, $D(x-y) = D(y-x)$. A Lorentz transformation can reverse the time order of spacelike-separated events, and D is Lorentz invariant (Peskin & Schroeder §2.4). So $C(x-y) = 0$



(a) retarded Δ_R (b) commutator C (c) Feynman Δ

Figure 30: Where each function is nonzero, relative to the light cone of y (dark: nonzero; white: zero; light: nonzero but decaying over a Compton wavelength). The retarded function lives only on and inside the future cone. The commutator lives inside both cones and vanishes at spacelike separation, which is microcausality. The Feynman function is nonzero everywhere.

outside the light cone: measurements at spacelike separation do not disturb each other, and Δ_R vanishes outside the future light cone (Fig. 30). No signal travels faster than light.

The Feynman function is an amplitude, not a signal. It does not vanish outside the light cone (Fig. 28). For $x^0 > y^0$ it is the amplitude for a particle created at y to be absorbed at x. For $x^0 < y^0$ it describes the reverse process, the antiparticle leg of §14.4; for a real field the particle is its own antiparticle.

The contrast with Part I. In the non-relativistic second-quantized theory $\hat{\psi}|0\rangle = 0$ kills the second term of the time-ordered product, so $\langle 0|\mathcal{T}\,\hat{\psi}\hat{\psi}^\dagger|0\rangle = \theta K$ and Feynman equals retarded (the remark at the end of §6.4). The relativistic field contains both $\hat{a}$ and $\hat{a}^\dagger$, so the second term survives. It is the price of antiparticles.

> **Caution: "causal" in the handwritten notes**
>
> The handwritten notes define $\langle x|y\rangle_{\text{causal}} = \langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle$ [hw. p. 24]. Taken literally, the overlap $\langle x|y\rangle = \langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle$ is the Wightman function. Inserting $\mathcal{T}$ is a definition: it takes the earlier point as the source. "Causal" there means time-ordered, that is, Feynman. It does not mean retarded.

16 Worldline equals field theory

16.1 The question [hw. pp. 22–23]

The handwritten notes ask how to show that the free relativistic particle of Chapters 10–13 and the free Klein–Gordon field of Chapter 15 are "physically equivalent" [hw. p. 22], and answer by showing that $K(x-y)$ is a constant times $\Delta(x-y)$ [hw. p. 23]. They give two methods: interpret K from the field side, and compute both sides and compare.

16.2 Method I: the amplitude from the field side [hw. pp. 23–25]

By §15.6, $|x\rangle = \hat{\phi}(x)|0\rangle$ creates a particle localized around $\mathbf{x}$ at time x^0, as well as a relativistic particle can be localized. It is the field-theory counterpart of Part I's $|x,t\rangle$ [hw. p. 37]. The amplitude for a particle created at y to be found at x is $\langle x|y\rangle = D(x-y)$ when $x^0 > y^0$. When $x^0 < y^0$ the physical process runs the other way, with amplitude $D(y-x)$. Taking the earlier point as the source in both cases gives

$$\langle x|y\rangle_{\text{causal}} \equiv \theta(x^0-y^0)D(x-y)+\theta(y^0-x^0)D(y-x) = \langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle = G_{\text{F}}(x-y) \tag{16.1}$$

[hw. p. 24]. The worldline theory was built from geometry and sums over every path, including the ones that turn back in time, so one expects K to be this function [hw. p. 25]. As §13.6 showed, this rests on the choice $T > 0$ together with $m^2 - i\eta$; it is not automatic.

16.3 Method II: explicit calculation [hw. pp. 25–26]

Work in Euclidean signature, where every integral converges. With the Euclidean propagator (13.14),

$$\bar{K}(\bar{x}-\bar{y}) = \int_0^\infty \mathrm{d}u\,\big(\alpha\pi\hbar u\big)^{-D/2}\exp\Big[-\frac{(\bar{x}-\bar{y})^2}{\alpha\hbar u}-\frac{u v^2}{\alpha\hbar}\Big], \tag{16.2}$$

take the Fourier transform in both arguments [hw. p. 25]:

$$\tilde{K}(\bar{p},\bar{q}) = \int \frac{\mathrm{d}^D\bar{x}}{(2\pi)^{D/2}}\frac{\mathrm{d}^D\bar{y}}{(2\pi)^{D/2}}\,e^{-i\bar{p}\cdot\bar{x}-i\bar{q}\cdot\bar{y}}\,\bar{K}(\bar{x}-\bar{y}). \tag{16.3}$$

The $\bar{x}$-integral is $J(\bar{p},u) = \int \mathrm{d}^D\bar{x}\,e^{-i\bar{p}\cdot\bar{x}-(\bar{x}-\bar{y})^2/\alpha\hbar u}$. Completing the square [hw. p. 26],

$$-\frac{(\bar{x}-\bar{y})^2}{\alpha\hbar u}-i\bar{p}\cdot\bar{x} = -\frac{1}{\alpha\hbar u}\Big(\bar{x}-\bar{y}+\frac{i\alpha\hbar u}{2}\bar{p}\Big)^2-\frac{\alpha\hbar u}{4}\bar{p}^2-i\bar{p}\cdot\bar{y}, \tag{16.4}$$

so $J = (\alpha\pi\hbar u)^{D/2}\, e^{-\alpha\hbar u\bar{p}^2/4 - i\bar{p}\cdot\bar{y}}$. Its prefactor cancels the one in $\bar{K}$. The $\bar{y}$-integral gives $\delta^D(\bar{p}+\bar{q})$, and the u-integral is elementary:

$$\tilde{K}(\bar{p},\bar{q}) = \delta^D(\bar{p}+\bar{q}) \int_0^\infty \mathrm{d}u\; e^{-u(\nu^2/\alpha\hbar + \alpha\hbar\bar{p}^2/4)} = \frac{\hbar\,\delta^D(\bar{p}+\bar{q})}{\frac{\alpha\hbar^2}{4}\bar{p}^2 + \frac{\nu^2}{\alpha}}. \tag{16.5}$$

Transforming back,

$$\bar{K}(\bar{x}-\bar{y}) = \hbar \int \frac{\mathrm{d}^D\bar{p}}{(2\pi)^D}\, \frac{e^{i\bar{p}\cdot(\bar{x}-\bar{y})}}{\frac{\alpha\hbar^2}{4}\bar{p}^2 + \frac{\nu^2}{\alpha}} = \frac{4}{\alpha\hbar}\int \frac{\mathrm{d}^D\bar{p}}{(2\pi)^D}\, \frac{e^{i\bar{p}\cdot(\bar{x}-\bar{y})}}{\bar{p}^2+\mu^2}, \qquad \mu = \frac{2\nu}{\alpha\hbar}. \tag{16.6}$$

The handwritten result on p. 26 carries an extra factor $-i$, the factor missing from §13.3. Comparing with the Euclidean field theory (15.9), $\bar{K} = \frac{4}{\alpha\hbar^2}\,(-i\bar{\Delta})$, and continuing back to real time, $K = \frac{4}{\alpha\hbar^2}G_\mathrm{F} = -\frac{4i}{\alpha\hbar^2}\Delta$.

16.4 The result

Worldline path integral = time-ordered two-point function of the field

$$K(x,y) = \frac{4}{\alpha\hbar^2}\,\langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle, \qquad \text{with the field's mass parameter} \quad \mu = \frac{m}{\hbar} = \frac{2\nu}{\alpha\hbar}. \tag{16.7}$$

With the handwritten choice $\alpha = 4/\hbar^2$ and $\nu^2 = 4\mu^2/\hbar^2$ [hw. p. 26],

$$K(x,y) = \langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle = \langle x|y\rangle_\mathrm{causal}. \tag{16.8}$$

Methods I and II agree exactly, with no leftover phase.

Why $\alpha = 4/\hbar^2$ is natural. With this value the worldline Hamiltonian (14.2) becomes $\hat{H} = (\hat{p}^2+m^2)/\hbar^2 = -\partial^2 + \mu^2$, the Klein–Gordon operator itself. Then (14.4) reads $K = \big\langle x\big|\frac{-i\hbar}{-\partial^2+\mu^2-i\eta}\big|y\big\rangle$, and by (15.8) this is $-i\Delta = G_\mathrm{F}$. The proper-time integral is literally Schwinger's representation of the inverse of the Klein–Gordon operator.

> **Caution: the "phase" and the units**
>
> The handwritten notes find $K = (\text{phase})\cdot\Delta$ [hw. pp. 23, 26]. With the handwritten prefactor the constant is $-\frac{4}{\alpha\hbar^2}$, a pure phase (-1) only for $\alpha = 4/\hbar^2$. With the factor i of §13.3 it is $-\frac{4i}{\alpha\hbar^2}$, which for $\alpha = 4/\hbar^2$ gives $K = -i\Delta = G_\mathrm{F}$ exactly. The conditions $\alpha = 4/\hbar^2$, $\nu^2 = 4m^2/\hbar^2$ of [hw. p. 26] are right once their m is read as $\mu = m/\hbar$, an inverse length. Only $\mu = 2\nu/\alpha\hbar$ is physical; α merely sets the overall constant. The rescaling $x \to \hbar x$ of [hw. p. 27] is not needed; it only compensates for writing the field side with $\hbar = 1$ in $e^{ip\cdot x}$ and in the mass. In units with $\hbar = 1$ the worldline action is the handwritten $S = \frac{1}{4}\int \mathrm{d}s\,[\lambda^{-1}\dot{z}^2 - 4\lambda\mu^2]$ [hw. p. 27].

16.5 What the equivalence means [hw. pp. 27–28]

The two-point function of the free field is a sum over the worldlines of one particle between the two points, over all proper durations $T > 0$, each weighted by $e^{iS/\hbar}$. The field's multi-particle states correspond to several independent worldlines (§17.6).

- **A matrix element of a larger theory.** In Part I the path integral computed a matrix element of Schrödinger's quantum mechanics (§7). Here the worldline integral computes a matrix element of a field theory, which also contains states with any number of particles [hw. p. 27].
- **From field theory back to the worldline.** Schwinger's parametrization (13.13) turns any propagator of a field-theory calculation into a proper-time integral, so field-theory results come out in worldline form [hw. pp. 27–28]. Only the gauge-invariant, T-integrated object can come out this way, since the einbein and its gauge freedom have no counterpart in the field theory.
- **Causality is not automatic.** It comes from the range $T > 0$ and the prescription $m^2 - i\eta$ (§13.6), not from the geometry alone ([hw. pp. 25, 28] say "automatically").

16.6 Why the worldline description is universal [hw. pp. 28–29]

The handwritten notes single out two ingredients of the worldline theory [hw. p. 28]. The first is *geometry*: the metric $\eta_{\mu\nu}$, or $G_{\mu\nu}$ in curved spacetime, and the einbein λ as a metric on the worldline, with tensor calculus under general coordinate transformations of both. The second is the *gauge procedure*. The successful match with field theory gives confidence in the formal gauge-fixing steps. In general relativity, the geodesic motion of a small body follows from the conservation of its energy–momentum tensor (the handwritten "EM tensor"), $\nabla_\mu T^{\mu\nu} = 0$.

Why should geometry know anything about the microscopic world? The handwritten answer is a stone [hw. p. 29]. A stone is some 10^{25} interacting quantum degrees of freedom, governed by causal fundamental interactions. At distances large compared with its size, its centre of mass nevertheless moves on a geodesic, whatever the microscopic details. The worldline theory is the universal long-distance description of any such object, and the Klein–Gordon field is its field-theory counterpart (this chapter). Spin, finite size and internal excitations enter as corrections suppressed by powers of size over distance, and the long-distance theory inherits its causality from the microscopic one. The same logic underlies modern worldline effective theories, which treat the black holes of an inspiralling binary as point particles with finite-size corrections.

17 Diagrams, background fields and interactions

In this chapter a line stands for $G_{\mathrm{F}} = \langle 0|\mathcal{T}\,\hat{\phi}\hat{\phi}|0\rangle$, which is the worldline amplitude K for $\alpha = 4/\hbar^2$ (16.8). The handwritten notes draw lines as $\Delta = iG_{\mathrm{F}}$; the factors of i are tracked below.

17.1 A line is a worldline path integral [hw. p. 29]

Draw the propagator as a line between its endpoints. A *Feynman rule* turns the drawing into the expression: a line from y to x means $G_{\mathrm{F}}(x-y)$. All drawings of a line with the same endpoints are topologically equivalent and mean the same thing [hw. p. 29]. The reason is now clear: the line *is* the worldline path integral, and its shape is integrated over. The lines of a Feynman diagram are not trajectories; each is a sum over all of them.

17.2 Insertions [hw. p. 30]

In Part I, inserting functions of the path into the path integral produced time-ordered operator products, eq. (7.5), with contact terms when velocities are inserted (§7.7). On a worldline the parameter s is unphysical, so a meaningful insertion must be reparametrization invariant. It is integrated along the worldline, $\int \mathrm{d}s\, F(x,\dot{x})$, with F of degree one in $\dot{x}$ (or carrying an einbein). Physically, insertions describe the interaction of the particle with *background fields* [hw. p. 30]. The handwritten notes leave the general language aside; the simplest case is enough to see how it works.

17.3 A charged particle in a background field: scalar QED [hw. pp. 31–33]

Scalar QED describes a scalar particle of mass m and charge q in an electromagnetic field. In a classical background $A_\mu(x)$, which is not integrated over [hw. p. 31],

$$K_A(x_f,x_i) = \int \frac{\mathcal{D}z\,\mathcal{D}\lambda}{V}\, e^{iS[z,\lambda;A]/\hbar}, \qquad S[z,\lambda;A] = S[z,\lambda] + S_{\mathrm{int}}[z;A]. \tag{17.1}$$

The interaction term. Consider the complete action $S[A] + S[z,\lambda] + S_{\mathrm{int}}$, where $S[A]$ lives on spacetime and $S[z,\lambda]$ on the worldline. Varying A_μ must give Maxwell's equations with the particle's current J^μ as the source, and J^μ does not depend on A. So $\delta S_{\mathrm{int}}/\delta A_\mu(x) = J^\mu(x)$, and S_{int} is linear in A [hw. p. 31]:

$$S_{\mathrm{int}} = \int \mathrm{d}^D x\, A_\mu(x)\, J^\mu(x), \qquad J^\mu(x) = q\int_0^1 \mathrm{d}s\, \dot{z}^\mu(s)\, \delta^D\big(x - z(s)\big), \tag{17.2}$$

so that

$$S_{\text{int}} = q \int_0^1 \mathrm{d}s\, A_\mu\big(z(s)\big)\, \dot{z}^\mu(s) = q \int A_\mu \, \mathrm{d}z^\mu \tag{17.3}$$

[hw. p. 32]. (The handwritten lines omit the $\int \mathrm{d}^D x$ and write the charge as e.)

- The current of a charge at rest is not zero: $J^0 = q\,\delta^N(\mathbf{x} - \mathbf{z}(t))$. A worldline always advances in z^0. The handwritten remark that the charge "must be moving to produce a current" applies only to the spatial current.
- S_{int} is reparametrization invariant without an einbein, because $A_\mu \dot{z}^\mu$ is of degree one in $\dot{z}$ (remark in §11.5).
- With $S[A] = -\frac{1}{4}\int F_{\mu\nu}F^{\mu\nu}$, the field equation is $\partial_\mu F^{\mu\nu} = -J^\nu$, which in mostly-plus conventions is Gauss's law $\nabla \cdot \mathbf{E} = \rho$ and its partners. The particle obeys the Lorentz force law, $m\,\mathrm{d}^2 z^\mu/\mathrm{d}\tau^2 = qF^\mu{}_\nu\,\mathrm{d}z^\nu/\mathrm{d}\tau$.

The quantum operator. The momentum is now $p_\mu = 2\dot{z}_\mu/(\alpha\lambda) + qA_\mu$, and the Hamiltonian (11.17) becomes $\lambda\frac{\alpha}{4}[(p - qA)^2 + m^2]$. Quantum mechanically this is $\frac{\alpha}{4}\big[-\hbar^2\mathcal{D}_\mu\mathcal{D}^\mu + m^2\big]$ with the covariant derivative $\mathcal{D}_\mu = \partial_\mu - \frac{iq}{\hbar}A_\mu$: the Klein–Gordon operator of scalar QED in a background field, minimal coupling.

Expanding in the charge. The factor $e^{iS_{\text{int}}/\hbar}$ expands into insertions [hw. p. 32]:

$$K_A = \int \frac{\mathcal{D}z\,\mathcal{D}\lambda}{V}\, e^{iS[z,\lambda]/\hbar}\Big[1 + \frac{iq}{\hbar}\int_0^1 \mathrm{d}s\, A_\mu \dot{z}^\mu + \frac{1}{2}\Big(\frac{iq}{\hbar}\Big)^2\Big(\int_0^1 \mathrm{d}s\, A_\mu \dot{z}^\mu\Big)^2 + \cdots\Big]. \tag{17.4}$$

(The handwritten line drops the $1/V$ in the second term.) The first term is the free propagator. Each further term attaches the background at a point of the worldline and integrates that point along the worldline (Fig. 31). The result is like a two-point function that includes every possible reaction with the background [hw. p. 33]. The insertions carry Lorentz indices, $\dot{z}^\mu$ contracted with A_μ: this is what the handwritten note on their "tensorial properties" refers to.

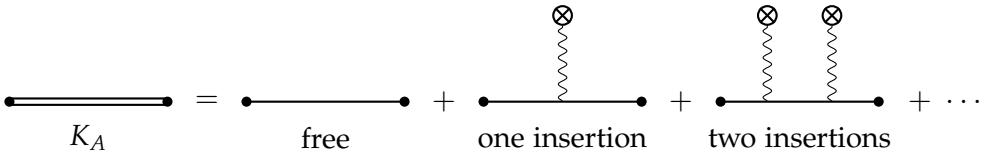


Figure 31: The propagator in a background field (double line) as a sum over insertions [hw. p. 32]. Each wavy line attaches the background (crossed circle) at a point of the worldline, integrated along it.

17.4 Gauge covariance and the Aharonov–Bohm phase★

The interaction enters only through the *Wilson line* $\mathcal{W}[z] = \exp\left(\frac{iq}{\hbar}\int_z A_\mu \, dz^\mu\right)$. Under a gauge transformation $A_\mu \to A_\mu + \partial_\mu \Lambda$, every path picks up the same change, $\int_z A\cdot dz \to \int_z A\cdot dz + \Lambda(x_f) - \Lambda(x_i)$, so

$$K_A(x_f,x_i) \;\to\; e^{iq\Lambda(x_f)/\hbar}\, K_A(x_f,x_i)\, e^{-iq\Lambda(x_i)/\hbar}. \tag{17.5}$$

The propagator transforms like $\psi(x_f)\psi^*(x_i)$ under $\psi \to e^{iq\Lambda/\hbar}\psi$. That is gauge covariance in one line; in field theory the same fact takes the Ward identities.

Two paths with the same endpoints differ by the phase $e^{(iq/\hbar)\oint A\cdot dz} = e^{iq\Phi/\hbar}$, where Φ is the magnetic flux through the loop they enclose. This is the Aharonov–Bohm effect, an exactly solvable non-relativistic example. Electrons passing on either side of a thin solenoid shift their interference pattern by $q\Phi/\hbar$, although $F_{\mu\nu} = 0$ everywhere along their paths. The sum over paths makes the effect obvious: the phase belongs to the path, not to the field at a point.

17.5 The seagull vertex from a contact term★

Scalar QED has a two-photon vertex, the "seagull", coming from the q^2A^2 in $(p - qA)^2 = p^2 - q(p\cdot A + A\cdot p) + q^2A^2$. On the worldline nobody puts it in by hand. It emerges from the second-order term of (17.4) through Part I's contact term.

In the gauge $\lambda = T$ with $s \in [0,1]$, the worldline action is $\int ds\, \frac{1}{2}m_w\dot z^2$ plus the mass term, with $m_w = 2/(\alpha T)$. Part I's result for velocity insertions, eq. (7.17), applies to every component, with the sign of its kinetic term:

$$\dot z^\mu(s)\,\dot z^\nu(s') \;\longrightarrow\; \mathcal{T}\big[\dot z^\mu(s)\dot z^\nu(s')\big] + \frac{i\hbar}{m_w}\,\eta^{\mu\nu}\,\delta(s-s'). \tag{17.6}$$

The contact part of the second-order term is

$$\frac{1}{2}\Big(\frac{iq}{\hbar}\Big)^2\frac{i\hbar}{m_w}\int_0^1 ds\, A_\mu A^\mu\big(z(s)\big) = -\frac{i\alpha T}{4\hbar}\int_0^1 ds\; q^2 A_\mu A^\mu\big(z(s)\big), \tag{17.7}$$

while the mass term in the exponent is $\frac{i}{\hbar}S_T \supset -\frac{iTv^2}{\alpha\hbar} = -\frac{i\alpha T}{4\hbar}\int_0^1 ds\; m^2$. The contact term therefore acts as the replacement $m^2 \to m^2 + q^2A_\mu A^\mu(z)$ along the path. That is exactly the seagull term (Fig. 32). It is the contact term of non-differentiable paths, eq. (4.14), showing up as a physical vertex.

17.6 Free N-point functions and Wick's theorem [hw. pp. 33–34]

Without backgrounds, the next objects are the N-point functions

$$G^{(N)}(x_1,\ldots,x_N) \equiv \langle 0|\mathcal{T}\,\hat\phi(x_1)\cdots\hat\phi(x_N)|0\rangle. \tag{17.8}$$

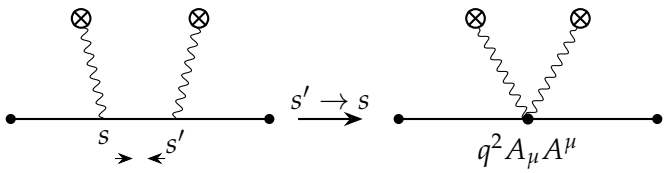


Figure 32: When two insertions coincide, the contact term $\frac{i\hbar}{m_w}\eta^{\mu\nu}\delta(s-s')$ of (17.6) produces a single vertex with two photons, the seagull vertex of scalar QED.

With lines alone, a diagram is a set of lines joining the external points in pairs.

- $G^{(N)} = 0$ for odd N, because each line has two ends [hw. p. 33].
- $G^{(4)}$ is a sum over the three pairings (Fig. 33):

$$\begin{aligned} G^{(4)} = {} & G_{\mathrm{F}}(x_1-x_2)G_{\mathrm{F}}(x_3-x_4) + G_{\mathrm{F}}(x_1-x_3)G_{\mathrm{F}}(x_2-x_4) \\ & + G_{\mathrm{F}}(x_1-x_4)G_{\mathrm{F}}(x_2-x_3). \end{aligned} \tag{17.9}$$

  In terms of $\Delta = iG_{\mathrm{F}}$ this is $-\big[\Delta_{12}\Delta_{34} + \Delta_{13}\Delta_{24} + \Delta_{14}\Delta_{23}\big]$. The handwritten products of Δ [hw. p. 33] miss this sign.

- In general $G^{(2n)}$ is the sum over all $(2n-1)!!$ pairings of products of G_{F}. This is *Wick's theorem*.

The crossing in the third pairing is an artefact of the drawing; the lines do not interact. Every $G^{(2n)}$ is built from $G^{(2)}$ alone: the particles are independent, and the theory has no self-interaction [hw. pp. 33–34]. Wick's theorem is the statement that the moments of a Gaussian are sums over pairings, the property of Gaussian integrals used throughout Part I. Part III derives it from the generating functional (§24.2).

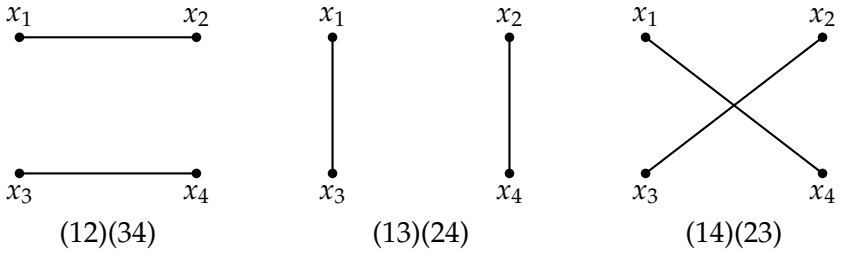


Figure 33: The three pairings that make up the free four-point function (17.9). They are topologically inequivalent; the crossing in the last one is only a feature of the drawing.

17.7 Closed worldlines: the vacuum loop★

$G^{(0)}$ has no external points, so its diagrams are closed worldlines. The handwritten notes write it as $\int \mathrm{d}^4x\, K(x-x)$ [hw. p. 34]. That is not quite right.

- $\int \mathrm{d}^D x\, K(x,x)$ describes a loop with a *marked point*, the point x where it starts and ends. On a closed loop of proper length T the marked point can sit anywhere, so this expression counts each loop T times over. The measure for closed loops is $\mathrm{d}T/T$. This is the residual rotation symmetry of §12.6.
- For a real field the two orientations of a loop are the same configuration, which gives a factor $\frac{1}{2}$ (Fig. 34).

In Euclidean signature, with $A = -\partial^2 + \mu^2$ and Frullani's integral $\int_0^\infty \frac{\mathrm{d}\sigma}{\sigma}(e^{-\sigma A} - e^{-\sigma}) = -\ln A$,

$$\ln Z_0 = \frac{1}{2}\int_0^\infty \frac{\mathrm{d}\sigma}{\sigma}\int \mathrm{d}^D\bar{x}\,(4\pi\sigma)^{-D/2}e^{-\mu^2\sigma} = -\frac{1}{2}\operatorname{Tr}\ln\big(-\partial^2+\mu^2\big) + \text{const.} \tag{17.10}$$

The marked-point expression is its derivative. Differentiating with respect to the mass marks a point on the loop,

$$\frac{\partial \ln Z_0}{\partial \mu^2} = -\frac{1}{2}\int \mathrm{d}^D\bar{x}\,\bar{G}(\bar{x},\bar{x}), \tag{17.11}$$

where $\bar{G} = (-\partial^2+\mu^2)^{-1}$, because the σ from differentiating $e^{-\mu^2\sigma}$ cancels the $1/\sigma$. So the handwritten $\int \mathrm{d}^4x\, K(x-x)$ is, up to constants, $\partial/\partial\mu^2$ of the vacuum loop. Both expressions diverge at short distance: $K(x,x)$ is the propagator at zero separation, and (17.10) diverges at $\sigma \to 0$. This is the vacuum energy: Part III computes it as the zero-point energy of the field (§24.4) and absorbs its divergence by renormalization (§30.4).

- **Part I connection.** A thermal trace is a sum over closed paths on a circle of circumference $\beta\hbar$ (§8.11). With periodic Euclidean time, closed worldlines can wind around that circle, and the winding loops give the free energy of a gas of relativistic particles.
- **Part III connection.** $-\frac{1}{2}\operatorname{Tr}\ln$ is the one-loop effective action (§29.2). In a constant background field it is the effective potential, this loop with μ^2 replaced by the curvature of the potential (§29.3). In an electromagnetic background (§17.3) the same loop gives the Euler–Heisenberg effective action and Schwinger's pair production in an electric field.

17.8 Interactions from vertices [hw. pp. 34–36]

To get connected $G^{(N)}$ with $N \geq 3$, the worldlines must be allowed to meet. Declare a set of *primitive diagrams*, the line and a vertex where three lines meet with strength g_3, and sum over all topologically inequivalent diagrams built from them [hw. p. 34]. For an interaction term $-\frac{g_3}{3!}\phi^3$ in the Lagrangian the exact rules, derived in Part III (§26.3), are:

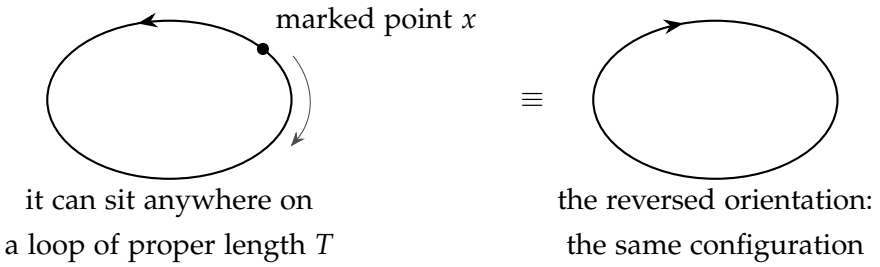


Figure 34: A closed worldline. Starting it at a marked point overcounts by the length T of the loop, hence $\mathrm{d}T/T$; the two orientations of a real field's loop are one configuration, hence $\frac{1}{2}$.

1. each line between points a and b gives $G_{\mathrm{F}}(a-b)$;
2. each vertex at y gives $-\frac{i}{\hbar}\,g_3\int \mathrm{d}^D y$;
3. divide by the symmetry factor of the diagram.

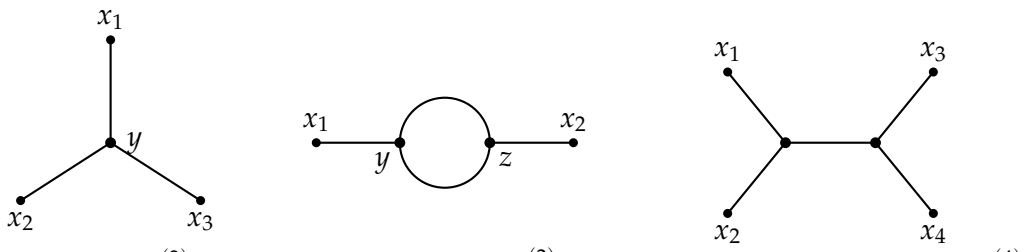
Some examples (Fig. 35). The handwritten versions on p. 35 are the same "up to overall factors" [hw. p. 36].

$$G_{\mathrm{c}}^{(3)} = -\frac{ig_3}{\hbar}\int \mathrm{d}^D y\; G_{\mathrm{F}}(x_1-y)G_{\mathrm{F}}(x_2-y)G_{\mathrm{F}}(x_3-y) + O(g_3^3), \tag{17.12}$$

$$G^{(2)} = G_{\mathrm{F}}(x_1-x_2) + \frac{1}{2}\Big(\frac{-ig_3}{\hbar}\Big)^2\int \mathrm{d}^D y\,\mathrm{d}^D z\; G_{\mathrm{F}}(x_1-y)\,G_{\mathrm{F}}(y-z)^2\,G_{\mathrm{F}}(z-x_2) + \cdots . \tag{17.13}$$

The $\frac{1}{2}$ in (17.13) is the symmetry factor of the bubble: its two internal lines can be exchanged. The dots include tadpoles, diagrams in which a loop hangs from a single line. At tree level $G^{(4)}$ is $(-ig_3/\hbar)^2$ times three exchange diagrams, one per channel, plus the disconnected pairings (17.9). The factors can be checked directly. To first order, $e^{(i/\hbar)\int \mathcal{L}_{\mathrm{int}}}$ contributes $-\frac{ig_3}{6\hbar}\int \mathrm{d}^D y\,\phi(y)^3$, and Wick's theorem contracts $\phi_1\phi_2\phi_3$ with $\phi(y)^3$ in $3! = 6$ equal ways, which gives (17.12). The subscript c marks the *connected* part. The remaining contractions join two external points to each other and the third to the vertex, whose other two legs close into a loop: at the same order they add a line times a *tadpole*, the one-point function of the cubic theory (§26.4).

Counting powers of $\hbar$. Each line carries one power of $\hbar$ (eq. (15.4)), and each vertex $1/\hbar$. A connected diagram with E external points, I internal lines, V vertices and $L = I - V + 1$ loops is therefore of order $\hbar^{E+I-V} = \hbar^{E-1+L}$. At fixed E each loop costs one power of $\hbar$: *the loop expansion is an expansion in $\hbar$*. It is the stationary-phase expansion of §4.6 in field-theory form, and Part III develops it (§20.5, §26.5).



(a) tree, $G^{(3)}$ (b) one loop, $G^{(2)}$ (c) one of three channels, $G^{(4)}$

Figure 35: Diagrams built from lines and cubic vertices [hw. p. 35]: the tree-level three-point function, the one-loop correction to the propagator, and one exchange channel of the four-point function.

The remarks of [hw. p. 36], completed.

1. $G^{(N)}$ receives contributions from infinitely many inequivalent diagrams, at all orders in the couplings.
2. Connected diagrams generate the disconnected ones.
3. The diagrams drawn are illustrations, not a complete list at a given order; the tadpoles are an example.
4. Loop integrals such as the bubble diverge at short distance in $D = 4$, since $G_{\mathrm{F}}(y - z)^2 \sim 1/|y - z|^4$. The "rigorous arguments" that cure this are regularization and renormalization (Chapter 30).
5. "A theory is its set of Feynman diagrams" [hw. p. 30] is a perturbative statement. Bound states and instantons are invisible at any finite order.

17.9 Summary: a theory is a propagator plus a topology [hw. pp. 36–37]

The main point of the handwritten discussion is that only two things define the perturbative theory [hw. p. 36]:

1. the *propagator*, given by the worldline path integral, which carries the mass;
2. the *topology* of the allowed diagrams, together with the strengths $g_3, g_4, \ldots$ of the vertices.

Spacetime field theory is an algorithm that produces the same $G^{(N)}$ with the same Feynman expressions [hw. p. 37]. Part III builds that algorithm from the generating functional $Z[J]$ (Chapters 23–27).

18 From particles to fields: the wave-functional picture

The handwritten Additional Note 03 [hw. pp. 37–40] sets up the Schrödinger representation of field theory: states as functionals of field configurations. It is placed last here because it is the bridge to Part III. Once field configurations have eigenstates, Part I's derivation of the path integral can be repeated for fields almost word for word (§18.5).

18.1 Two analogies, not one [hw. p. 37]

The note starts from the analogy oscillator $\to$ Klein–Gordon field: $\hat{x}(t) \to \hat{\tilde{\phi}}(t, \mathbf{k})$, $|t, \mathbf{x}\rangle \to \hat{\phi}(x)|0\rangle$, and in the relativistic context $\langle y|x\rangle = \langle 0|\mathcal{T}\,\hat{\phi}(y)\hat{\phi}(x)|0\rangle$ [hw. p. 37]. Two different analogies are mixed in this list.

	first-quantized picture (Chapters 14–16)	second-quantized picture (this chapter)
what plays "x"	the particle's position $\mathbf{x}$, a label of $\hat{\phi}(x)$	the oscillator coordinate: the field amplitude $\phi(\mathbf{x})$ at each $\mathbf{x}$
analogue of $\|x\rangle$	the one-particle state $\hat{\phi}(x)\|0\rangle$	the field eigenstate $\|[\phi]\rangle$
analogue of $\psi(x)$	a Klein–Gordon wavefunction	a wave functional $\Psi[\phi]$

Both are correct, but they answer different questions. The question of [hw. p. 37], which state in field theory is the analogue of $|t, \mathbf{x}\rangle$, has two answers. It is $\hat{\phi}(x)|0\rangle$ if $\mathbf{x}$ is read as the particle's position (Chapter 16). It is $|t, [\phi]\rangle$ if the oscillator's coordinate is meant, as in this chapter. The oscillator's x is the field amplitude, not a point in space (Fig. 36).

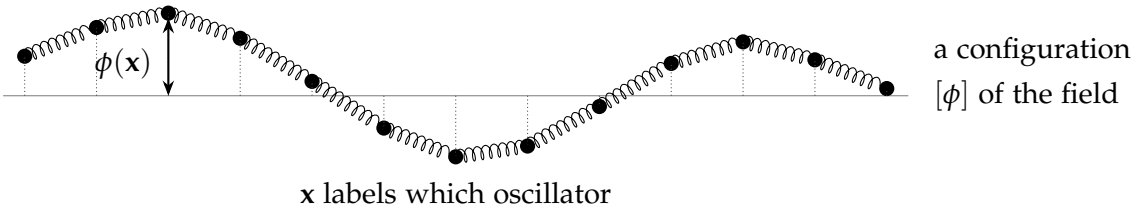


Figure 36: A free field as a mattress of coupled oscillators. Each site $\mathbf{x}$ carries an oscillator whose coordinate is the field amplitude $\phi(\mathbf{x})$, shown as the vertical displacement; the springs are the gradient energy $\frac{1}{2}(\nabla\phi)^2$. A field eigenstate $|[\phi]\rangle$ fixes all the displacements at once, as $|x\rangle$ fixes one oscillator's.

18.2 Field eigenstates [hw. pp. 37–39]

In the Schrödinger picture, take simultaneous eigenstates of the field operator at every point,

$$\hat{\phi}(\mathbf{x})|[\phi]\rangle = \phi(\mathbf{x})|[\phi]\rangle \quad \text{for all } \mathbf{x}, \qquad |t,[\phi]\rangle = e^{i\hat{H}t/\hbar}|[\phi]\rangle, \tag{18.1}$$

the second with Part I's convention for Heisenberg states. The label is a whole function $\phi(\mathbf{x})$, a point in the space of field configurations, just as a path was a point in path space (Fig. 2). The handwritten notes use the momentum modes, $\hat{\tilde{\phi}}(\mathbf{k})|[\tilde{\phi}]\rangle = \tilde{\phi}(\mathbf{k})|[\tilde{\phi}]\rangle$ [hw. p. 38]. Since $\hat{\tilde{\phi}}(\mathbf{k})^\dagger = \hat{\tilde{\phi}}(-\mathbf{k})$, these operators are not Hermitian. They commute, so simultaneous eigenstates exist, but the eigenvalues must satisfy $\tilde{\phi}(-\mathbf{k}) = \tilde{\phi}(\mathbf{k})^*$: they are the Fourier transforms of a real $\phi(\mathbf{x})$. The Hermitian $\hat{\phi}(\mathbf{x})$ of (18.1) avoids the issue. (The handwritten label "position-space representation" means this field-amplitude representation, the analogue of an oscillator's position representation.)

Orthonormality and completeness [hw. p. 38].

$$\langle[\phi]|[\phi']\rangle = \delta[\phi-\phi'] \equiv \prod_{\mathbf{x}} \delta\big(\phi(\mathbf{x})-\phi'(\mathbf{x})\big) = \int \mathcal{D}\kappa \, \exp\Big[i\int \mathrm{d}^N x\, \kappa(\mathbf{x})\big(\phi(\mathbf{x})-\phi'(\mathbf{x})\big)\Big], \tag{18.2}$$

$$\int \mathcal{D}\phi \, |[\phi]\rangle\langle[\phi]| = \mathbb{1}, \qquad |\Psi\rangle = \int \mathcal{D}\phi \, \Psi[\phi]\,|[\phi]\rangle, \qquad \Psi[\phi] = \langle[\phi]|\Psi\rangle. \tag{18.3}$$

The functional delta imposes equality at every point. Here $\mathcal{D}\kappa = \prod_{\mathbf{x}} \mathrm{d}\kappa(\mathbf{x})/2\pi$, one factor $1/2\pi$ per mode (the handwritten measure has $(2\pi)^N$). All of this is defined on a spatial lattice, where $\prod_{\mathbf{x}}$ is a finite product, with the continuum as a limit, exactly as for paths in Part I (§2). $\Psi[\phi]$ is the *wave functional* [hw. p. 39].

18.3 The momentum as a functional derivative [hw. pp. 39–40]

The canonical commutator $[\hat{\phi}(\mathbf{x}), \hat{\Pi}(\mathbf{y})] = i\hbar\,\delta^N(\mathbf{x}-\mathbf{y})$ [hw. p. 39] is represented, as $\langle x|\hat{p}|x'\rangle = -i\hbar\,\partial_x\delta(x-x')$ was in Part I, with the functional derivative of §2:

$$\langle[\phi]|\hat{\Pi}(\mathbf{x})|\Psi\rangle = -i\hbar\,\frac{\delta\Psi[\phi]}{\delta\phi(\mathbf{x})}. \tag{18.4}$$

In momentum space $[\hat{\tilde{\phi}}(\mathbf{k}), \hat{\tilde{\Pi}}(\mathbf{q})] = i\hbar\,\delta^N(\mathbf{k}+\mathbf{q})$ (Appendix C), so $\hat{\tilde{\Pi}}(-\mathbf{k})$ is conjugate to $\hat{\tilde{\phi}}(\mathbf{k})$ and acts as $-i\hbar\,\delta/\delta\tilde{\phi}(\mathbf{k})$ [hw. p. 40]. The Hamiltonian density of (15.1) is $\frac{1}{2}\Pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}\mu^2\phi^2$, so the Schrödinger equation for wave functionals is

$$i\hbar\,\partial_t\Psi[\phi,t] = \int \mathrm{d}^N x\,\Big[-\frac{\hbar^2}{2}\frac{\delta^2}{\delta\phi(\mathbf{x})^2} + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}\mu^2\phi^2\Big]\Psi[\phi,t]. \tag{18.5}$$

This is the "Schrödinger representation of QFT" of [hw. p. 37]: formal in the continuum, well defined on a lattice.

18.4 The vacuum wave functional★

The exercise of [hw. p. 40] asks for the vacuum wave functional $\Psi_0[\phi] = \langle[\phi]|0\rangle$ and its normalization. At $t = 0$ the mode expansions of Chapter 15 invert to

$$\hat{a}_{\mathbf{k}} = \sqrt{\frac{\omega_{\mathbf{k}}}{2\hbar}}\,\hat{\tilde{\phi}}(\mathbf{k}) + \frac{i}{\sqrt{2\hbar\omega_{\mathbf{k}}}}\,\hat{\tilde{\Pi}}(\mathbf{k}). \tag{18.6}$$

With $\hat{\tilde{\Pi}}(\mathbf{k}) \to -i\hbar\,\delta/\delta\tilde{\phi}(-\mathbf{k})$, the condition $\hat{a}_{\mathbf{k}}|0\rangle = 0$ becomes

$$\omega_{\mathbf{k}}\,\tilde{\phi}(\mathbf{k})\,\Psi_0 + \hbar\,\frac{\delta\Psi_0}{\delta\tilde{\phi}(-\mathbf{k})} = 0. \tag{18.7}$$

Its solution is a Gaussian:

$$\boxed{\Psi_0[\tilde{\phi}] = \mathcal{N}\exp\Big[-\frac{1}{2\hbar}\int \mathrm{d}^N k\;\omega_{\mathbf{k}}\,|\tilde{\phi}(\mathbf{k})|^2\Big]} \tag{18.8}$$

since $\delta/\delta\tilde{\phi}(-\mathbf{k})$ of $\int \mathrm{d}^N q\,\omega_{\mathbf{q}}\tilde{\phi}(\mathbf{q})\tilde{\phi}(-\mathbf{q})$ is $2\omega_{\mathbf{k}}\tilde{\phi}(\mathbf{k})$.

- **One oscillator ground state per mode.** Each factor is $\psi_0(u) \propto e^{-\omega u^2/2\hbar}$, the ground state of a unit-mass oscillator (§9). The vacuum is the product of the ground states of all the oscillators of §15.1.
- **Normalization.** $\langle 0|0\rangle = \int \mathcal{D}\phi\,|\Psi_0|^2 = 1$ gives, formally, $\mathcal{N} = \prod(\omega_{\mathbf{k}}/\pi\hbar)^{1/4}$, one factor per real mode. Each pair $\pm\mathbf{k}$ carries two real modes, the real and imaginary parts of $\tilde{\phi}(\mathbf{k})$. The product is defined on a lattice in a box.
- **In position space** the exponent is $-\frac{1}{2\hbar}\int \mathrm{d}^N x\,\mathrm{d}^N y\;\phi(\mathbf{x})\,E(\mathbf{x}-\mathbf{y})\,\phi(\mathbf{y})$, with kernel $E(\mathbf{x}) = \int \frac{\mathrm{d}^N k}{(2\pi)^N}\,\omega_{\mathbf{k}}\,e^{i\mathbf{k}\cdot\mathbf{x}}$. The kernel is non-local: for $\mu > 0$ it extends over a Compton wavelength. The Gaussian's covariance is $\langle 0|\hat{\phi}(\mathbf{x})\hat{\phi}(\mathbf{y})|0\rangle = \frac{\hbar}{2}E^{-1}(\mathbf{x}-\mathbf{y}) = D(\mathbf{x}-\mathbf{y},0)$, the equal-time overlap (15.12). The vacuum correlates neighbouring regions of space, the seed of the entanglement between regions that underlies the Unruh effect and black-hole entropy.

18.5 Toward Part III: the field path integral★

With field eigenstates in hand, the derivation of Part I, Chapter 5, goes through with $x \to \phi(\mathbf{x})$ and $p \to \Pi(\mathbf{x})$. Slice $e^{-i\hat{H}(t_f-t_i)/\hbar}$ into small steps, insert the completeness relation (18.3) between them, and insert momentum eigenstates inside each step. The Hamiltonian is quadratic in Π, so the momentum integrals are Gaussian (Case 1, §5.3), and the result is

$$\langle[\phi_f]|e^{-i\hat{H}(t_f-t_i)/\hbar}|[\phi_i]\rangle = \int_{\phi(t_i)=\phi_i}^{\phi(t_f)=\phi_f}\mathcal{D}\phi\;e^{iS_{\mathrm{KG}}[\phi]/\hbar}, \tag{18.9}$$

with the Klein–Gordon action (15.1), $S_{\mathrm{KG}} = \int \mathrm{d}^D x\,\big[-\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}\mu^2\phi^2\big]$. Part III starts from this formula (§19.3).

Bridge to Part III: what carries over

- **The measure.** $\mathcal{D}\phi$ is Part I's time slicing applied at every point of space at once (§19.4).
- **The propagator.** Gaussian functional integrals give G_{F} (Chapter 15, §24.1), and Wick's theorem (§17.6, §24.2) is the statement that their moments are sums over pairings.
- $i\eta$ **and Euclidean time.** The same prescription and rotations as §13.4 and §15.5 (§23.4). Euclidean field theory is the statistical mechanics of fields (§8.7, §23.5).
- **One loop.** The determinant of the Gaussian is the closed worldline, $-\frac{1}{2}\,\mathrm{Tr}\ln$ (§17.7, §29.3).
- **Interactions.** Expanding $e^{iS_{\mathrm{int}}/\hbar}$ reproduces the diagram rules of §17.8 (Chapter 26), and the loop expansion is the expansion in $\hbar$.
- **Gauge fields.** Dividing by the gauge volume needs the Faddeev–Popov procedure of §12.6, now with ghost fields. Gauge fields lie beyond this volume.

C Solutions to the exercises

The handwritten notes pose several "show that" exercises. Their solutions are collected here. The vacuum wave functional of [hw. p. 40] is solved in §18.4.

C.1 The oscillator Green's function [hw. p. 21]

With $\hat{x}(t) = \sqrt{\hbar/2\omega}\,(\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t})$ and $\hat{a}|0\rangle = 0$, $\langle 0|\hat{x}(t)\hat{x}(t')|0\rangle = \frac{\hbar}{2\omega}e^{-i\omega(t-t')} \equiv D(t-t')$. Hence $\Delta(t) = i\big[\theta(t)D(t) + \theta(-t)D(-t)\big] = \frac{i\hbar}{2\omega}e^{-i\omega|t|}$. Differentiating, with $\partial_t|t| = \operatorname{sgn} t$ and $\partial_t \operatorname{sgn} t = 2\delta(t)$:

$$\partial_t\Delta = \frac{\hbar}{2}\operatorname{sgn}(t)\,e^{-i\omega|t|}, \qquad \partial_t^2\Delta = \hbar\,\delta(t) - \frac{i\hbar\omega}{2}e^{-i\omega|t|}, \tag{C.1}$$

and since $\omega^2\Delta = \frac{i\hbar\omega}{2}e^{-i\omega|t|}$,

$$\big(\partial_t^2 + \omega^2\big)\Delta(t) = \hbar\,\delta(t). \tag{C.2}$$

This is Part I's contact term, eq. (7.11), with $m = 1$, multiplied by i and evaluated in the vacuum.

C.2 The Klein–Gordon Green's function [hw. p. 21]

From (15.4) and (15.5), renaming $\mathbf{k} \to -\mathbf{k}$ in the term with $\theta(y^0 - x^0)$,

$$\Delta(x-y) = \int \frac{\mathrm{d}^N k}{(2\pi)^N}\; e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}\;\frac{i\hbar}{2\omega_{\mathbf{k}}}e^{-i\omega_{\mathbf{k}}|x^0-y^0|}. \tag{C.3}$$

The operator $-\partial^2 + \mu^2 = \partial_t^2 - \nabla^2 + \mu^2$ acts on each mode as $\partial_t^2 + \omega_{\mathbf{k}}^2$, since $-\nabla^2 \to \mathbf{k}^2$. By the previous result each mode gives $\hbar\,\delta(x^0 - y^0)$, and $\int \frac{\mathrm{d}^N k}{(2\pi)^N}e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})} = \delta^N(\mathbf{x}-\mathbf{y})$. So $(-\partial^2 + \mu^2)\Delta(x-y) = \hbar\,\delta^D(x-y)$.

C.3 The energy and momentum representations [hw. p. 22]

The integrand of (15.7) has poles where $E^2 = \omega^2 - i\eta$, at $E_\pm = \pm(\omega - i\eta')$ with $\eta' = \eta/2\omega$. The pole E_+ lies just below the real axis and E_- just above it. For $t > 0$, e^{-iEt} decays in the lower half-plane. Closing there (clockwise) picks up E_+, with residue $e^{-i\omega t}/2\omega$, so $\int \frac{\mathrm{d}E}{2\pi}\frac{e^{-iEt}}{E^2-\omega^2+i\eta} = -\frac{i}{2\omega}e^{-i\omega t}$. For $t < 0$, closing in the upper half-plane (counterclockwise) around E_-, with residue $-e^{i\omega t}/2\omega$, gives $-\frac{i}{2\omega}e^{i\omega t}$. Multiplying by $-\hbar$ gives $\Delta(t) = \frac{i\hbar}{2\omega}e^{-i\omega|t|}$, as above. For the field, apply this to each mode with $\omega \to \omega_{\mathbf{k}}$ and $E = k^0$. In mostly-plus signature $E^2 - \omega_{\mathbf{k}}^2 + i\eta = -(k^2 + \mu^2 - i\eta)$, which gives (15.8).

C.4 The Euclidean representation [hw. p. 22]

Start from (15.8) and rotate the k^0 contour counterclockwise onto the imaginary axis, $k^0 = i\bar{k}^0$. The rotation sweeps the first and third quadrants, while the poles sit in the second and fourth (Fig. 29), and the arcs vanish because the integrand falls like $1/|k^0|^2$. Then $\mathrm{d}k^0 = i\,\mathrm{d}\bar{k}^0$ and $k^2 = (\bar{k}^0)^2 + \mathbf{k}^2$. Continuing $x^0 = -i\bar{x}^0$ makes $k^0x^0 = \bar{k}^0\bar{x}^0$, so $k\cdot x = -\bar{k}^0\bar{x}^0 + \mathbf{k}\cdot\mathbf{x}$. Relabelling $\bar{k}^0 \to -\bar{k}^0$ in the even integrand gives the Euclidean product $\bar{k}\cdot\bar{x}$, and (15.9) follows, the factor i coming from $\mathrm{d}k^0$.

C.5 Lorentz invariance of the normalization [hw. p. 23]

Since $\delta\big(\omega_{\mathbf{k}}^2 - (k^0)^2\big) = \big[\delta(k^0 - \omega_{\mathbf{k}}) + \delta(k^0 + \omega_{\mathbf{k}})\big]/2\omega_{\mathbf{k}}$,

$$\int \mathrm{d}^D k\; \delta(k^2 + \mu^2)\,\theta(k^0)\,f(k) = \int \frac{\mathrm{d}^N k}{2\omega_{\mathbf{k}}}\, f(\omega_{\mathbf{k}}, \mathbf{k}). \tag{C.4}$$

The left side is Lorentz invariant: $\mathrm{d}^D k$ and k^2 are invariant, and so is the sign of k^0 for a timelike k under orthochronous transformations. So $\mathrm{d}^N k/2\omega_{\mathbf{k}}$ is an invariant measure, and $2\omega_{\mathbf{k}}\delta^N(\mathbf{k}-\mathbf{q})$ is the delta function with respect to it: $\int \frac{\mathrm{d}^N k}{2\omega_{\mathbf{k}}}\, 2\omega_{\mathbf{k}}\delta^N(\mathbf{k}-\mathbf{q}) f(\mathbf{k}) = f(\mathbf{q})$. Hence $\langle \mathbf{k}|\mathbf{q}\rangle = 2\omega_{\mathbf{k}}\delta^N(\mathbf{k}-\mathbf{q})$ takes the same value in every frame, which is the statement $\omega_{\mathbf{k}}\delta(\mathbf{k}-\mathbf{q}) = \omega_{\mathbf{k}'}\delta(\mathbf{k}'-\mathbf{q}')$ of [hw. p. 23].

C.6 The commutator in momentum space [hw. p. 39]

With $\tilde{\phi}(\mathbf{k}) = \int \frac{\mathrm{d}^N x}{(2\pi)^{N/2}} e^{-i\mathbf{k}\cdot\mathbf{x}}\phi(\mathbf{x})$, and the same for $\tilde{\Pi}$,

$$\begin{aligned}[\hat{\tilde{\phi}}(\mathbf{k}), \hat{\tilde{\Pi}}(\mathbf{q})] &= \int \frac{\mathrm{d}^N x\,\mathrm{d}^N y}{(2\pi)^N}\, e^{-i\mathbf{k}\cdot\mathbf{x} - i\mathbf{q}\cdot\mathbf{y}}\, i\hbar\,\delta^N(\mathbf{x}-\mathbf{y}) \\ &= i\hbar \int \frac{\mathrm{d}^N x}{(2\pi)^N}\, e^{-i(\mathbf{k}+\mathbf{q})\cdot\mathbf{x}} = i\hbar\,\delta^N(\mathbf{k}+\mathbf{q}).\end{aligned} \tag{C.5}$$

The same follows from the mode expansions at $t = 0$, using $[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{q}}] = \delta^N(\mathbf{k}-\mathbf{q})$:

$$[\hat{\tilde{\phi}}(\mathbf{k}), \hat{\tilde{\Pi}}(\mathbf{q})] = -\frac{i\hbar}{2}\sqrt{\frac{\omega_{\mathbf{q}}}{\omega_{\mathbf{k}}}}\,\big([\hat{a}_{\mathbf{k}}, -\hat{a}^\dagger_{-\mathbf{q}}] + [\hat{a}^\dagger_{-\mathbf{k}}, \hat{a}_{\mathbf{q}}]\big) = i\hbar\,\delta^N(\mathbf{k}+\mathbf{q}), \tag{C.6}$$

which is a check on the normalization of the dictionary in §15.1.

C.7 The retarded Green's function

For one mode, $[\hat{x}(t), \hat{x}(0)] = \frac{\hbar}{2\omega}\big(e^{-i\omega t} - e^{i\omega t}\big) = -\frac{i\hbar}{\omega}\sin\omega t$, so

$$\Delta_{\mathrm{R}}(t) = i\theta(t)\langle 0|[\hat{x}(t), \hat{x}(0)]|0\rangle = \theta(t)\,\frac{\hbar}{\omega}\sin\omega t. \tag{C.7}$$

Write $\Delta_{\mathrm{R}} = \theta g$ with $g(0) = 0$ and $g'(0) = \hbar$. Using eq. (2.10), $\delta' g = -g'(0)\delta$, so

$$\partial_t^2(\theta g) = \delta' g + 2\delta g' + \theta g'' = \hbar\,\delta(t) - \omega^2 \theta g \quad \Longrightarrow \quad \big(\partial_t^2 + \omega^2\big)\Delta_{\mathrm{R}} = \hbar\,\delta(t). \tag{C.8}$$

Mode by mode, as in the Klein–Gordon case above, $(-\partial^2 + \mu^2)\Delta_{\mathrm{R}} = \hbar\,\delta^D$. Δ_{R} and Δ are two Green's functions of the same operator, with different boundary conditions.

D Map of the handwritten notes and correction log

D.1 Where each handwritten page went

pages	content	here
1–2	paths to worldlines; the internal parameter	§10
3	proper time, geodesics, the square-root action	§10.4, §11.1
4	Euclidean sign; equations of motion; problems	§11.1–11.3
5–7	the einbein action; reparametrization invariance	§11.4–11.5
7–8	recap; ν and the mass; the gauge volume	§11.1, §12.2
8–9	physical and gauge symmetries	§12.1
10–11	orbits and gauge slices	§12.3, §12.5
12–16	the constant gauge; Fourier construction; M	§12.4, §12.6
16–18	working formula; proper-time gauge; Gaussian	§13.1–13.3
19–20	four approaches; oscillator $\leftrightarrow$ Klein–Gordon	§10.2, §15.1
20–21	mode expansions, commutators, Green's functions	§15.1–15.3
22	energy, momentum and Euclidean representations	§14.1, §15.4–15.5
23–24	one-particle states; $\langle x\|y\rangle_{\text{causal}}$	§15.6, §16.2
25–27	explicit calculation and matching; comments	§16.3–16.5
28–29	Schwinger parametrization; geometry; universality	§16.5–16.6, §17.1
30	insertions	§17.2
31–33	scalar QED in a background; free N-point functions	§17.3, §17.6
34–36	vacuum diagram; interactions; remarks	§17.7–17.8
36–37	summary: propagator plus topology	§17.9
37–40	Additional Note 03: Schrödinger picture of QFT	§18

D.2 Corrections

pp. 1–2 Position eigenstates $|x^\mu\rangle$ live in an enlarged Hilbert space and are not physical states. The amplitude written $\langle x_f|x_i\rangle$ is $\langle x_f|\hat{G}|x_i\rangle$ (§14.2). Proper time parametrizes only timelike paths; the sum also contains spacelike pieces and pieces that run backward in time (§10.3).

p. 3 Timelike geodesics *maximize* proper time, not "max or min" (§10.4). The boundary condition is $x(1) = x_f$, not $x(s) = x_f$.

pp. 3, 4, 7 "$\mathrm{d}\tau/\mathrm{d}s = 0$: massless" is not a branch of the equations of motion. The square-root action has no massless limit (null helix); the einbein action handles $m = 0$ (§11.2, §11.4).

p. 5 The positive root $\lambda = \sqrt{-\dot{x}^2}/v$ is selected (§11.4).

p. 6 The factor $\frac{1}{2}$ of p. 5 is dropped; harmless.

p. 8 "$v \propto$ mass" made exact: $m = 2v/\alpha$; the Compton wavelength sets the quantum scale (§11.1).

p. 9 The einbein is not needed to make reparametrization invariance manifest. It makes an invariant action quadratic in x and valid at $m = 0$ (§11.4).

p. 10 The path $x^\mu(s)$ *can* be gauge-fixed (static gauge), but not globally: that gauge drops the paths that turn back in time (§12.5).

p. 11 A good slice must meet every orbit (existence) as well as meet each at most once (§12.3).

pp. 12–13 The endpoint values $\lambda(0) = a$, $\lambda(1) = b$ are not gauge invariant and cannot be fixed. $\lambda > 0$ is required, so $e \in (0, \infty)$, not $\mathbb{R}$. "$f(s) = 1$" means the identity map $f(s) = s$ (§12.4).

pp. 14–15 A series in $\{1, \cos 2\pi ns, \sin 2\pi ns\}$ always has $X(0) = X(1)$; non-periodic functions are represented only in the mean-square sense. The Fourier construction is replaced by the one-line argument $T = \int \lambda \, \mathrm{d}s$ and half-range modes (§12.4, §12.6).

p. 16 Upper limit s in $Y(s)$; the modulus runs over $(0, \infty)$. $M(e) = 1$ is justified (§12.6).

p. 17 The e-dependence is not eliminated by the rescaling; it moves into the length of the interval (§13.2).

p. 18 The time component of the Gaussian has the conjugate Fresnel phase, giving an overall factor i; no second Wick rotation is needed (§13.3). The prescription $v^2 \to v^2 - i\eta$ is added (§13.4).

p. 20 $\hat{H}_{\mathrm{HO}} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$, not $\hbar\omega(\hat{a}\hat{a}^\dagger + \frac{1}{2})$. The field Hamiltonian needs $\hbar$ and is normal-ordered (§15.1).

p. 21 The Wightman function needs $\hbar/2\omega_{\mathbf{k}}$ (§15.3).

p. 22 $K_r = \theta(t - t_0)\langle x, t | y, t_0 \rangle$.

pp. 23, 26 "$K = (\text{phase})\,\Delta$": in general $K = \frac{4}{\alpha\hbar^2} G_\text{F}$; with $\alpha = 4/\hbar^2$, $K = G_\text{F} = -i\Delta$ exactly (§16.4).

p. 24 $\hat\phi(x)|0\rangle$ carries a factor $\sqrt{\hbar}$. The states $\hat\phi(0,\mathbf{x})|0\rangle$ are not orthogonal (§15.6). The literal overlap $\langle x|y\rangle$ is the Wightman function, and "causal" means Feynman, not retarded (§15.7).

pp. 25, 28 Causality is not automatic: it needs $T > 0$ and $m^2 - i\eta$ (§13.6).

pp. 25–26 The extra $-i$ in $\tilde K$ traces back to p. 18 (§16.3).

pp. 26–27 The field-side "m" is $\mu = m/\hbar$; only $\mu = 2\nu/\alpha\hbar$ is physical. The rescaling $x \to \hbar x$ is not needed (§16.4).

p. 28 "EM tensor" is the energy–momentum tensor (§16.6).

pp. 31–32 $S_\text{int} = \int \mathrm{d}^D x\, A_\mu J^\mu$; it is linear in A because $\delta S_\text{int}/\delta A_\mu = J^\mu$; a charge at rest has $J^0 \neq 0$ (§17.3).

p. 32 The second term of the expansion is missing its $1/V$.

p. 33 With $\Delta = iG_\text{F}$ the free four-point function is $-(\Delta_{12}\Delta_{34} + \Delta_{13}\Delta_{24} + \Delta_{14}\Delta_{23})$ (§17.6).

p. 34 The vacuum diagram is not $\int \mathrm{d}^4 x\, K(x-x)$, which is a loop with a marked point (the μ^2-derivative of the vacuum loop). Closed loops need $\mathrm{d}T/T$ and $\frac{1}{2}$ (§17.7).

pp. 35–36 Exact factors for the cubic diagrams (§17.8). "Rigorous arguments" means regularization and renormalization; the diagrammatic definition of a theory is perturbative.

p. 37 Two analogies, particle position and oscillator coordinate, are separated (§18.1).

p. 38 The momentum modes are not Hermitian, so their eigenvalues obey $\tilde\phi(-\mathbf{k}) = \tilde\phi(\mathbf{k})^*$. The measure of the functional delta is $\mathrm{d}\kappa/2\pi$ per mode (§18.2).

p. 40 The Klein–Gordon action needs parentheses; the exercise is solved (§18.4).

notation Couplings $\lambda, \lambda_3, \lambda_4 \to g_3, g_4$; charge $e \to q$; modulus $e \to T$; field-side $p \to k$ and $m \to \mu$; $i\epsilon \to i\eta$.